

# MODULAR KOSZUL DUALITY AND THE SHADOW TOWER: ALGEBRAICITY, DEPTH CLASSIFICATION, AND THE GRAVITATIONAL COPRODUCT

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**ABSTRACT.** Let  $\mathcal{A}$  be a chirally Koszul vertex algebra with modular characteristic  $\kappa$ , cubic shadow  $\alpha$ , and quartic shadow  $S_4$ . We prove four results. First, the weighted shadow generating function  $H(t) = \sum_{r \geq 2} r S_r t^r$  satisfies  $H(t)^2 = t^4 Q(t)$  where  $Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$  is the *shadow metric*: the tower is algebraic of degree 2, and every  $S_r$  for  $r \geq 5$  is determined by the three seeds  $(\kappa, \alpha, S_4)$ . Second, the critical discriminant  $\Delta = 8\kappa S_4$  partitions the standard landscape into four depth classes: Gaussian ( $r_{\max} = 2$ ), Lie ( $r_{\max} = 3$ ), contact ( $r_{\max} = 4$ ), and mixed ( $r_{\max} = \infty$ ). Third, the genus-2 free energy is  $F_2 = 7\kappa/5760$  for single-generator algebras; the first multi-channel computation for  $\mathcal{W}_3$  exhibits a cross-channel correction  $\delta F_2 = (c+204)/(16c)$ , resolving the multi-generator universality question negatively. Fourth, the gravitational coproduct obtained by BRST reduction from  $\widehat{\mathfrak{sl}}_2$  to Virasoro is strictly primitive at all degrees; the mechanism applies to all principal Drinfeld–Sokolov reductions.

*Remark 0.1* (Relationship to companion papers). The  $c$ -independent identity  $S_3(\text{Vir}_c) = 2$  is now proved as a theorem; see [16]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and GZ26 commuting Hamiltonians is developed in [17].

## 1. INTRODUCTION

**1.1. The question.** Classical Koszul duality is a phenomenon of graded algebras over a field. A quadratic algebra  $\mathcal{A}$  determines a dual coalgebra  $\mathcal{A}^!$ ; the bar construction  $B(\mathcal{A})$  mediates between them; and the Koszul property (bar cohomology concentrated in bar degree 1) is equivalent to the existence of a linear resolution. The theory is the subject of Priddy [15], Beilinson–Ginzburg–Soergel [2], and Loday–Vallette [13].

On an algebraic curve  $X$ , the generators of the algebra become sections of a  $\mathcal{D}_X$ -module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic 1-form  $\eta_{ij} = d \log(z_i - z_j)$  on the Fulton–MacPherson compactification  $\overline{C}_n(X)$  of the configuration space. The resulting object, the *chiral bar complex*  $\tilde{B}^{\text{ch}}(\mathcal{A})$ , is a factorization coalgebra on the Ran space  $\text{Ran}(X)$  [1, 4, 6].

At genus 0, the classical theory embeds via restriction to the formal disk (the embedding requires specifying a retraction  $\mathbb{A}^1 \rightarrow \text{pt}$  and its homotopy transfer; the  $E_1/E_\infty$  bar distinction, the Arnol’d algebra of  $\text{Conf}_n(\mathbb{A}^1)$ , and the Fulton–MacPherson compactifications are already absent over a bare point). At genus  $g \geq 1$ , the topology of the curve forces further structure: the bar differential acquires curvature  $d_{\text{fb}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$ , where  $\omega_g = c_1(\mathbb{E})$  is the Hodge class on  $\overline{\mathcal{M}}_g$  and the scalar  $\kappa(\mathcal{A}) \in \mathbb{Q}(c)$  is an intrinsic invariant of the algebra called the *modular characteristic*. This curvature has no analogue in the classical setting; its geometric origin is the nontriviality of the Hodge bundle over  $\overline{\mathcal{M}}_g$ .

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The present paper addresses a single question: *what invariants does the chiral bar construction produce, beyond the modular characteristic?*

**1.2. The answer: the shadow obstruction tower.** The bar complex on a curve of genus  $g$  with  $n$  marked points defines a pronilpotent dg Lie algebra  $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$  (the modular convolution algebra), whose differential encodes graph sums over stable curves. The boundary-of-boundary relation  $\partial^2 = 0$  on  $\overline{\mathcal{M}}_{g,n}$  implies  $D^2 = 0$  on  $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ , and the element  $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$  is therefore a Maurer–Cartan element:  $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ .

The MC element  $\Theta_{\mathcal{A}}$  encodes the full genus expansion. Its finite-order projections  $\Theta_{\mathcal{A}}^{\leq r}$ , obtained by truncating the degree filtration at level  $r$ , form the *shadow obstruction tower*. At degree 2, the tower produces the modular characteristic  $\kappa$ . At degree 3, a cubic shadow  $\alpha$  (the structure constant of the transferred  $L_{\infty}$  bracket  $\ell_3$ ). At degree 4, a quartic shadow  $S_4$  (the first genuinely nonlinear invariant, measuring the contact term in the four-point collision residue).

The obstruction to truncating  $\Theta_{\mathcal{A}}$  at finite degree is measured by classes  $o_{r+1}(\mathcal{A}) \in H^2(\mathcal{A}_{r+1,0}^{\text{sh}})$  in the shadow algebra  $\mathcal{A}^{\text{sh}} := H_{\bullet}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$ . When  $o_{r+1} = 0$ , the tower stops; when  $o_{r+1} \neq 0$ , the next shadow coefficient  $S_{r+1}$  is forced. The question is how far this process continues and what controls it.

**1.3. The shadow metric and algebraicity.** A single quadratic polynomial controls the entire shadow obstruction tower (Theorem 4.1, parts (i)–(ii)).

Restrict to a one-dimensional primary slice  $L$  of the cyclic deformation complex, spanned by a single primary field  $\eta$  of definite conformal weight. On  $L$ , the MC recursion reduces to a scalar recursion for the shadow coefficients  $S_r$ , and the weighted generating function  $H(t) = \sum_{r \geq 2} r S_r t^r$  satisfies

$$H(t)^2 = t^4 Q(t),$$

where the *shadow metric*

$$Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$$

is a quadratic polynomial determined by the three seed invariants  $(\kappa, \alpha, S_4)$ . The generating function is algebraic of degree 2:  $H(t) = t^2 \sqrt{Q(t)}$ , and every shadow coefficient  $S_r$  for  $r \geq 5$  follows from the binomial expansion of  $\sqrt{Q}$ .

The proof converts the Maurer–Cartan recursion into a convolution identity for the Taylor coefficients of  $F(t) = H(t)/t^2$ , showing  $F^2 = Q$ . Orders  $m = 0, 1, 2$  reproduce the definitions of  $\kappa, \alpha, S_4$ ; orders  $m \geq 3$  are the recursion.

This is the *intrinsic quartic principle*: three numbers determine an infinite sequence of invariants. The analogue is the Weierstrass equation: two parameters  $(g_2, g_3)$  determine all Laurent coefficients of  $\wp$ , because  $\wp$  satisfies an algebraic equation of degree 2 in  $\wp'$ . Here three parameters  $(\kappa, \alpha, S_4)$  determine all  $S_r$ , because  $H$  satisfies an algebraic equation of degree 2.

**1.4. The four-class classification.** Write  $Q(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$  for the *Gaussian decomposition*, where  $\Delta := 8\kappa S_4$  is the *critical discriminant*. The Gaussian envelope  $(2\kappa + 3\alpha t)^2$  is the part of  $Q$  visible to the quadratic and cubic shadows alone;  $\Delta$  measures the irreducible quartic contribution.

The critical discriminant controls a dichotomy.

When  $\Delta = 0$ , the shadow metric is a perfect square:  $Q = (2\kappa + 3\alpha t)^2$ ,  $H(t) = t^2(2\kappa + 3\alpha t)$  is polynomial, and the tower terminates. If  $\alpha = 0$ , only  $S_2 = \kappa$  survives (depth 2); if  $\alpha \neq 0$ , then  $S_2$  and  $S_3$  survive (depth 3).

When  $\Delta \neq 0$ , the shadow metric is irreducible:  $\sqrt{Q}$  is irrational, every Taylor coefficient beyond degree 1 carries a factor of  $\Delta$ , and  $S_r \neq 0$  for all  $r \geq 4$  generically. The tower is infinite.

Every chirally Koszul algebra whose primary-line shadow data has been computed falls into one of four *shadow depth classes* (Theorem 4.1, part (iii)). The partition is exhaustive for the standard Lie-theoretic landscape (Table 1).

| Class               | $\Delta$ | $r_{\max}$ | Prototype                 | Shadow data                          |
|---------------------|----------|------------|---------------------------|--------------------------------------|
| <b>G</b> (Gaussian) | 0        | 2          | Heisenberg                | $\kappa = k$                         |
| <b>L</b> (Lie)      | 0        | 3          | Affine Kac–Moody          | $\kappa, \alpha \neq 0$              |
| <b>C</b> (Contact)  | 0*       | 4          | $\beta\gamma$ system      | $\kappa = 6\lambda^2 - 6\lambda + 1$ |
| <b>M</b> (Mixed)    | $\neq 0$ | $\infty$   | Virasoro, $\mathcal{W}_N$ | $\kappa, \alpha, S_4$                |

\* Class **C**:  $\Delta = 0$  on the primary line; the quartic contact invariant lives on a charged stratum.

Class **G**: the bar complex is classically formal, the tower terminates at  $\kappa$ , and all  $F_g$  follow from  $\kappa$  alone. Class **L**: a single cubic correction (the Lie structure constant) appears, after which the Jacobi identity kills the tower. Class **C**: the quartic contact invariant  $S_4$  is nonzero but lives on a separate stratum whose self-bracket exits the complex, terminating the tower one step later. Class **M**: the tower is infinite,  $F_g$  receives corrections at every degree, and the shadow coefficients  $S_r$  decay or grow at a rate governed by the *shadow growth rate*

$$\rho(\mathcal{A}) = \sqrt{\frac{9\alpha^2 + 16\kappa S_4}{4\kappa^2}}.$$

The growth rate equals the reciprocal of the branch-point modulus. When  $\rho < 1$ , the tower converges absolutely; when  $\rho > 1$ , it diverges, but  $\sqrt{Q}$  extends analytically past the convergence radius. For Virasoro at central charge  $c$ :

$$\rho(\text{Vir}_c) = \frac{1}{c} \sqrt{\frac{180c + 872}{5c + 22}},$$

with a critical central charge  $c_\star \approx 6.125$  (the unique positive root of  $5c^3 + 22c^2 - 180c - 872 = 0$ ) separating the convergent and divergent regimes. At the self-dual point  $c = 13$  (where the Koszul involution  $\text{Vir}_c \mapsto \text{Vir}_{26-c}$  has a fixed point, so  $\text{Vir}_{13}$  is self-dual under uncurved quadratic Koszul duality):  $\rho \approx 0.467$ .

Classical Koszul duality over a field embeds into the genus-0, degree-2,  $\Delta = 0$  stratum (the Gaussian case where the tower terminates immediately) via restriction to the formal disk; the embedding is not an equivalence, since the configuration-space geometry and the  $E_1/E_\infty$  distinction are already absent over a bare point. Everything beyond class **G** is new.

**1.5. The shadow connection and Koszul monodromy.** The shadow metric equips each primary line with a logarithmic connection

$$\nabla^{\text{sh}} = d - \frac{Q'(t)}{2Q(t)} dt,$$

with regular singularities at the zeros of  $Q$  and residue  $\frac{1}{2}$ . The monodromy around each branch point is  $-1$ . This is the *Koszul sign*: the square root  $\sqrt{Q}$  changes sign around a simple zero, and the shadow obstruction tower inherits this sign change through the algebraic relation  $H = t^2 \sqrt{Q}$ .

Under Koszul duality  $\mathcal{A} \mapsto \mathcal{A}^!$ , the shadow metric transforms by the Verdier involution. For the Virasoro family  $\text{Vir}_c^! = \text{Vir}_{26-c}$ , the discriminants satisfy

$$\Delta(\text{Vir}_c) + \Delta(\text{Vir}_{26-c}) = \frac{6960}{(5c + 22)(152 - 5c)},$$

a Virasoro-specific identity whose numerator is  $6960 = 2^4 \cdot 3 \cdot 5 \cdot 29$ . The dual singularities  $c = -22/5$  and  $c = 152/5$  sum to 26: they are Koszul partners. Other families (affine Kac–Moody,  $\mathcal{W}_N$  with  $N \geq 3$ ) produce a different numerator.

**1.6. Genus-2 free energies.** The shadow obstruction tower produces invariants of moduli spaces through graph sums over stable curves. The genus- $g$  free energy  $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$  holds for single-generator (uniform-weight) chirally Koszul algebras at all genera, and unconditionally at genus 1 for all families. Here  $\lambda_g^{\text{FP}} = \int_{\overline{\mathcal{M}}_{g,1}} \psi_1^{2g-2} \lambda_g$  is the Faber–Pandharipande integral, with  $\lambda_g = c_g(\mathbb{E})$  the top Chern class of the Hodge bundle and  $\psi_1$  the cotangent line class.

At genus 2:  $\lambda_2^{\text{FP}} = 7/5760$ , and  $F_2(\mathcal{A}) = \kappa \cdot 7/5760$  for every single-generator chirally Koszul algebra. Verification requires an explicit sum over the seven stable graphs of genus 2 with no marked points (smooth genus-2 curve, figure-eight, banana, dumbbell, theta, tadpole, and chain), weighted by automorphism factors and vertex contributions from the shadow obstruction tower.

For the  $\mathcal{W}_3$  algebra, which has two generators  $T$  (weight 2) and  $W$  (weight 3) producing two primary lines, the genus-2 computation is a multi-channel graph sum: each edge of the stable graph carries a channel label ( $T$  or  $W$ ), each vertex contributes a shadow coefficient on the corresponding primary line, and the propagator  $P_T = 2/c$ ,  $P_W = 3/c$  depends on the channel. The per-channel diagonal sum yields  $F_2^{\text{scal}}(\mathcal{W}_3) = \kappa \cdot \lambda_2^{\text{FP}} = 7c/6912$ , but the full multi-channel graph sum produces an additional cross-channel correction  $\delta F_2 = (c + 204)/(16c)$ . The multi-generator universality problem is resolved negatively: the cross-channel correction is not cancelled by CohFT  $R$ -matrix terms, so  $F_g \neq \kappa \cdot \lambda_g^{\text{FP}}$  at genus  $\geq 2$  for multi-weight algebras. The correct decomposition is  $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$ .

The Maurer–Cartan equation on  $\overline{\mathcal{M}}_{2,0}$  also produces a *planted-forest correction*

$$\delta_{\text{pf}}^{(2,0)} = \frac{S_3(10S_3 - \kappa)}{48},$$

measuring the contribution of codimension-2 boundary strata to the genus-2 amplitude. This correction vanishes for class **G** (where  $S_3 = 0$ ). For classes **C** and **M**, it is generically nonzero. For the Virasoro algebra:  $\delta_{\text{pf}}^{(2,0)} = -(c - 40)/48$ .

**1.7. The gravitational coproduct.** The fourth result concerns BRST reduction. Drinfeld–Sokolov (DS) reduction takes affine Kac–Moody data to  $\mathcal{W}$ -algebra data via a BRST complex with ghosts; the homological perturbation lemma (HPL) transfers the  $\mathcal{A}_\infty$  products and coproduct from the affine dg-shifted Yangian to a transferred Yangian on the  $\mathcal{W}$ -algebra sector.

We prove the result for  $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$ . The ghost-number obstruction (the homotopy  $h$  has ghost degree  $-1$ ; the projection  $p \otimes p$  kills all ghost-number- $(-1)$  elements) forces the transferred coproduct to be *strictly primitive at all degrees*. The mechanism applies to any principal DS reduction  $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\widehat{\mathfrak{g}})$ , since the ghost-number grading is universal; we state the general case as a conjecture and prove it for  $\widehat{\mathfrak{sl}}_2$ :

$$\Delta_z^{\text{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T, \quad \Delta_{z,n}^{\text{Vir}} = 0 \quad \text{for } n \geq 2.$$

The ghost-number mechanism kills all higher-degree coproduct corrections. The only nontrivial gravitational structure is the  $r$ -matrix twist  $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$ , encoding the braiding of line operators.

The physical content: in Chern–Simons theory, gluons split (the Clebsch–Gordan decomposition makes the coproduct nontrivial). In 3d gravity, gravitons do not split: the coproduct is primitive, and all interactions reside in the twist.

**1.8. Relation to existing work.** The bar construction for chiral algebras was introduced by Beilinson and Drinfeld [1] and developed in the factorization algebra framework by Francis and Gaitsgory [6]

and Costello and Gwilliam [4]. The modular characteristic  $\kappa$  appears implicitly in the work of Feigin and Fuchs on Hodge bundles of conformal blocks, and the formula  $\kappa(\text{Vir}_c) = c/2$  can be extracted from Beilinson and Schechtman [3]. The invariants of moduli spaces produced by vertex algebras via conformal blocks are the subject of Faber and Pandharipande [7], Mumford [14], and the extensive literature on cohomological field theories following Kontsevich and Manin [10].

The quartic shadow  $S_4$ , the critical discriminant  $\Delta$ , the four-class partition, and the algebraicity theorem (Theorem 4.1) appear to be new. The connection between shadow depth and  $L_\infty$  non-formality, while consonant with the general philosophy of Kontsevich [11], has not been established previously in the chiral setting.

The Drinfeld–Sokolov reduction and its homological algebra are classical [5, 8, 12]; the gravitational coproduct primitivity result (Theorem 9.2) and its connection to the shadow obstruction tower appear to be new.

*Remark 1.1* (Relationship to companion papers). The Virasoro spectral R-matrix  $R(z) = z^{2h} \exp(-(c/4)/z^2)$  and its connection to the Riccati shadow structure are developed in the companion paper *The Virasoro spectral R-matrix* (virasoro\_r\_matrix.tex).

## 2. THE SHADOW METRIC AND THE RICCATI EQUATION

**2.1. The shadow obstruction tower.** Let  $\mathcal{A}$  be a chirally Koszul vertex algebra with modular cyclic deformation complex  $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$  and universal Maurer–Cartan element  $\Theta_{\mathcal{A}} \in \text{MC}(\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A}))$ . The complex carries a degree filtration  $F^\bullet$  with  $F^r = \bigoplus_{s \geq r} \text{Def}_{\text{cyc}}^{(s)}$ , and the *shadow obstruction tower* is the sequence of finite-order truncations

$$\Theta_{\mathcal{A}}^{\leq 2}, \Theta_{\mathcal{A}}^{\leq 3}, \Theta_{\mathcal{A}}^{\leq 4}, \dots$$

where  $\Theta_{\mathcal{A}}^{\leq r}$  solves the Maurer–Cartan equation in the quotient  $\mathfrak{g}_{\mathcal{A}}^{\text{amb}}/F^{r+1}$ . The degree- $r$  projection  $\text{Sh}_r(\mathcal{A})$  is the  $r$ -th *shadow* of  $\mathcal{A}$ . On a one-dimensional slice of the deformation complex,  $\text{Sh}_r$  is determined by a single scalar  $S_r$ ; the MC equation projected to degree  $r$  gives the *all-degree master equation*

$$\nabla_H(\text{Sh}_r) + \mathfrak{o}^{(r)} = 0, \quad (2.1)$$

where  $\nabla_H$  is the linearized operator and  $\mathfrak{o}^{(r)}$  is the obstruction from lower degrees.

### 2.2. Primary lines and the shadow metric.

**Definition 2.1** (Primary line). A *primary line* in  $\text{Def}_{\text{cyc}}^{\text{mod}}(\mathcal{A})$  is a one-dimensional subspace  $L = \mathbf{k} \cdot \eta$  spanned by a single primary field  $\eta$  of definite conformal weight  $h_\eta$ . For a chiral algebra with generators  $\phi_1, \dots, \phi_N$  of weights  $h_1, \dots, h_N$ , each generator spans a primary line; the full first-order deformation space is  $\bigoplus_i L_i$ .

Let  $L$  be a primary line with coordinate  $x$ , and write  $\text{Sh}_r|_L = S_r x^r$  for the shadow data restricted to  $L$ . Set

$$\kappa := S_2, \quad \alpha := S_3.$$

The scalar  $\kappa$  is the modular characteristic (curvature) of  $\mathcal{A}$  on  $L$ ; the scalar  $\alpha$  is the cubic shadow.

**Definition 2.2** (Shadow metric). Define the *weighted initial data*  $a_0 := 2\kappa$ ,  $a_1 := 3\alpha$ ,  $a_2 := 4S_4$ . The *shadow metric* (shadow quadratic form) on  $L$  is the polynomial

$$Q_L(t) := a_0^2 + 2a_0 a_1 t + (a_1^2 + 2a_0 a_2) t^2 = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2 \in k(c)[t]. \quad (2.2)$$

The *critical discriminant* of  $L$  is

$$\Delta := 8\kappa S_4 = a_0 a_2.$$

*Remark 2.3.* The shadow metric  $Q_L$  is a quadratic polynomial in  $t$ , determined by three OPE invariants:  $\kappa, \alpha, S_4$ . The term “metric” refers to  $Q_L$ ’s role as the quadratic form governing the tower’s geometry (branch points, monodromy, convergence radius), not to a Riemannian metric.

**2.3. The recursion on a primary line.** On a primary line  $L$ , the all-degree master equation (2.1) reduces to a recursion among the scalars  $S_r$ .

**Proposition 2.4** (Single-line recursion). *On a primary line  $L$  with propagator  $P := 1/\kappa$ , the MC equation at degree  $r \geq 5$  is*

$$S_r = -\frac{P}{2r} \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \quad c_{jk} = \begin{cases} 1 & \text{if } j < k, \\ 1/2 & \text{if } j = k. \end{cases} \quad (2.3)$$

*This recursion is well-defined: since  $j \geq 3$  and  $j + k = r + 2$ , every term on the right has  $k = r + 2 - j \leq r - 1$ , so  $S_r$  is determined by  $\{S_2, S_3, S_4, \dots, S_{r-1}\}$ . The degree-4 shadow  $S_4$  is an intrinsic OPE invariant (the quartic contact term in the four-point collision residue), not determined by the recursion; it serves as initial data alongside  $\kappa = S_2$  and  $\alpha = S_3$ .*

*Proof.* The Lie bracket on the cyclic deformation complex, restricted to  $L$ , pairs  $\text{Sh}_j = S_j x^j$  and  $\text{Sh}_k = S_k x^k$  to produce a term of degree  $j + k - 2$  with coefficient  $jk P S_j S_k$ . The eigenvalue identity  $\nabla_H(x^r) = 2r x^r$  separates the MC equation at degree  $r$  into the linear term  $2r S_r$  on the left and the obstruction sum on the right, giving the stated recursion. The symmetry factor  $c_{jk}$  accounts for the factor  $\frac{1}{2}$  in  $\frac{1}{2}[\Theta, \Theta]$  when  $j = k$ .  $\square$

**2.4. The weighted shadow generating function.**

**Definition 2.5** (Weighted shadow generating function). *The weighted shadow generating function on  $L$  is*

$$H(t) := \sum_{r \geq 2} r S_r t^r.$$

Setting  $a_n := (n + 2) S_{n+2}$  for  $n \geq 0$ , this becomes  $H(t) = \sum_{n \geq 0} a_n t^{n+2}$ , with initial values  $a_0 = 2\kappa$ ,  $a_1 = 3\alpha$ ,  $a_2 = 4S_4$ .

The weighted generating function  $H$  packages the recursion (2.3) as a convolution identity. Set  $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$ . The MC recursion forces every Cauchy coefficient of  $F^2$  beyond degree 2 to vanish, so  $F^2$  is a polynomial of degree 2.

### 3. ALGEBRAICITY AND THE DEPTH CLASSIFICATION

**3.1. The Riccati algebraicity theorem.**

**Theorem 3.1** (Riccati algebraicity). *Let  $L$  be a primary line with shadow data  $S_r$ , shadow metric  $Q_L$  (Definition 2.2), and weighted shadow generating function  $H$  (Definition 2.5). Then the all-degree master equation on  $L$  is equivalent to the algebraic relation*

$$H(t)^2 = t^4 Q_L(t). \quad (3.1)$$

*The shadow generating function is therefore algebraic of degree 2 over  $k(\kappa, \alpha, S_4)(t)$ :*

$$H(t) = t^2 \sqrt{Q_L(t)}.$$

*Proof.* Set  $F(t) := H(t)/t^2 = \sum_{n \geq 0} a_n t^n$  where  $a_n = (n + 2) S_{n+2}$ . The claim (3.1) is equivalent to  $F(t)^2 = Q_L(t)$ . We verify this by computing the Cauchy product  $F^2 = \sum_{m \geq 0} (\sum_{i+j=m} a_i a_j) t^m$  and showing it agrees with  $Q_L$  at all orders.

*Orders 0, 1, 2 (initial data).* At  $m = 0$ :  $a_0^2 = (2\kappa)^2 = 4\kappa^2$ . At  $m = 1$ :  $2 a_0 a_1 = 2 \cdot 2\kappa \cdot 3\alpha = 12\kappa\alpha$ . At  $m = 2$ :  $a_1^2 + 2 a_0 a_2 = 9\alpha^2 + 2 \cdot 2\kappa \cdot 4S_4 = 9\alpha^2 + 16\kappa S_4$ . These are exactly the coefficients of  $Q_L(t)$  in (2.2).

*Orders  $m \geq 3$  (the MC recursion forces vanishing).* Since  $Q_L$  has degree 2 in  $t$ , the identity  $F^2 = Q_L$  requires

$$\sum_{i+j=m} a_i a_j = 0 \quad \text{for all } m \geq 3. \quad (3.2)$$

We show that this is exactly the MC recursion (2.3) at degree  $r = m + 2$ .

The sum  $\sum_{i+j=m} a_i a_j$  with  $a_i = (i+2) S_{i+2}$  and  $a_j = (j+2) S_{j+2}$  equals  $\sum_{p+q=m+4, p,q \geq 2} p q S_p S_q$ , where  $p = i+2$ ,  $q = j+2$ . Setting  $r = m+2$ , this becomes  $\sum_{p+q=r+2, p,q \geq 2} p q S_p S_q$ .

Isolating the terms with  $p = 2$  or  $q = 2$  (which contribute  $2\kappa$  from  $S_2 = \kappa$ ):

$$\begin{aligned} \sum_{p+q=r+2, p,q \geq 2} p q S_p S_q &= 2 \cdot 2r \cdot \kappa S_r + \sum_{\substack{p+q=r+2 \\ p,q \geq 3}} p q S_p S_q \\ &= 4r\kappa S_r + 2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k, \end{aligned}$$

where the factor 2 in front of the restricted sum accounts for  $S_p S_q + S_q S_p$  when  $p \neq q$ , and  $c_{jk}$  corrects for the diagonal. Substituting the recursion (2.3):

$$2 \sum_{\substack{j+k=r+2 \\ 3 \leq j \leq k}} c_{jk} j k S_j S_k = 2 \cdot (-2r\kappa S_r) = -4r\kappa S_r.$$

Hence  $\sum_{p+q=r+2} p q S_p S_q = 4r\kappa S_r - 4r\kappa S_r = 0$ , which is (3.2).

Conversely, the vanishing (3.2) at order  $m \geq 3$  implies the recursion (2.3) at  $r = m+2$  by solving for  $S_r$  (using  $\kappa \neq 0$ ).

The equivalence of (3.1) with the full MC recursion is therefore established at all degrees. Taking positive square roots (the branch with  $F(0) = a_0 = 2\kappa > 0$ ) gives  $H(t) = t^2 F(t) = t^2 \sqrt{Q_L(t)}$ .  $\square$

*Remark 3.2 (Riccati ODE reformulation).* Set  $U(t) := \sum_{r \geq 3} S_r t^r$  (the nonlinear part, removing  $\text{Sh}_2$ ). The algebraic relation (3.1) is equivalent to the Riccati-type first-order ODE

$$2t U'(t) + \frac{P}{2} (U'(t))^2 = R_3 t^3 + R_4 t^4,$$

where  $R_3 = 6\alpha$  and  $R_4 = (9\alpha^2 + 2\Delta)/(2\kappa)$  encode the cubic and quartic inputs. The left-hand side is quadratic in  $U'$ : this is the precise sense in which the shadow obstruction tower is algebraic of degree 2.

### 3.2. Gaussian decomposition.

**Corollary 3.3** (Gaussian decomposition). *The shadow metric decomposes as*

$$Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2,$$

where  $\Delta = 8\kappa S_4$  is the critical discriminant. The first term  $(2\kappa + 3\alpha t)^2$  is the Gaussian envelope: the shadow metric of the truncated tower with  $S_r = 0$  for  $r \geq 4$ . The second term  $2\Delta t^2$  is the interaction correction, measuring the quartic OPE contribution not captured by the cubic data alone.

*Proof.*  $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha t + 9\alpha^2 t^2$ . Adding  $2\Delta t^2 = 16\kappa S_4 t^2$  recovers the coefficient  $9\alpha^2 + 16\kappa S_4$  of  $t^2$  in  $Q_L(t)$ .  $\square$

### 3.3. Intrinsic quartic principle.

**Corollary 3.4** (Intrinsic quartic principle). *On a primary line  $L$ , the shadow obstruction tower has intrinsic OPE data at degrees 2, 3, 4 only. All shadow coefficients  $S_r$  for  $r \geq 5$  are determined by  $(\kappa, \alpha, S_4)$  alone, via the closed form*

$$S_r = \frac{1}{r} [t^{r-2}] \sqrt{Q_L(t)} \quad (r \geq 2).$$

*Proof.*  $Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$  depends on exactly  $(\kappa, \alpha, S_4)$ . Since  $H(t) = t^2 \sqrt{Q_L(t)} = \sum_{r \geq 2} r S_r t^r$  by Theorem 3.1, comparing coefficients gives  $r S_r = [t^{r-2}] \sqrt{Q_L}$ .  $\square$

### 3.4. The shadow connection.

**Proposition 3.5** (Shadow connection). *The shadow metric defines a logarithmic connection on  $\mathcal{O}_L$  over  $L \setminus \{Q_L = 0\}$ :*

$$\nabla^{\text{sh}} = d - \omega, \quad \omega := \frac{Q'_L(t)}{2Q_L(t)} dt = d(\log \sqrt{Q_L}).$$

- (i) *The connection is flat:  $(\nabla^{\text{sh}})^2 = 0$ .*
- (ii) *At each simple zero  $t_0$  of  $Q_L$ , the residue of  $\omega$  is  $\frac{1}{2}$ .*
- (iii) *The monodromy is  $\exp(2\pi i \cdot \frac{1}{2}) = -1$ : the standard monodromy of a square root around a simple branch point.*
- (iv) *The flat section with  $\Phi(0) = 1$  is  $\Phi(t) = \sqrt{Q_L(t)/Q_L(0)} = \sqrt{Q_L(t)/(2\kappa)}$ , and the shadow generating function is  $H(t) = 2\kappa t^2 \Phi(t)$ : the parallel transport of the curvature along the degree parameter, weighted by  $t^2$ .*

*Proof.* Flatness is automatic:  $\omega$  is a closed 1-form on a 1-dimensional base. At a simple zero  $t_0$  of  $Q_L$ :  $Q_L(t) \approx Q'_L(t_0)(t - t_0)$ , so  $\omega \approx \frac{1}{2(t-t_0)} dt$ , giving residue  $\frac{1}{2}$ . The monodromy exponentiation is immediate. For (iv):  $\nabla^{\text{sh}}\Phi = 0$  since  $d\Phi = \omega\Phi$  by construction, and the evaluation at  $t = 0$  gives  $\Phi(0) = \sqrt{Q_L(0)/(2\kappa)} = 2\kappa/(2\kappa) = 1$ .  $\square$

*Remark 3.6.* The shadow generating function  $H(t) = 2\kappa t^2 \Phi(t)$  is *not* a flat section of  $\nabla^{\text{sh}}$ : the factor  $t^2$  contributes  $\nabla^{\text{sh}}(H) \neq 0$ . The flat section is  $\Phi(t) = \sqrt{Q_L(t)/(2\kappa)}$ ; the shadow obstruction tower is the Taylor expansion of  $t^2 \cdot \Phi(t)$ , reflecting the degree offset (the tower starts at degree 2, not degree 0).

### 3.5. The single-line dichotomy.

**Theorem 3.7** (Single-line dichotomy). *Let  $L, S_r, \alpha, \Delta, Q_L$  be as above, and define the shadow depth  $r_{\max}|_L := \max\{r : S_r \neq 0\}$  (with  $r_{\max} = \infty$  if  $S_r \neq 0$  for infinitely many  $r$ ). Then  $r_{\max}|_L \in \{2, 3, \infty\}$ , classified by the shadow metric:*

- (i) **Class G** ( $\Delta = 0, \alpha = 0$ ).  $Q_L = (2\kappa)^2$  is constant.  $H(t) = 2\kappa t^2$ . Only  $S_2$  survives:  $r_{\max} = 2$ .
- (ii) **Class L** ( $\Delta = 0, \alpha \neq 0$ ).  $Q_L = (2\kappa + 3\alpha t)^2$  is a perfect square.  $H(t) = t^2(2\kappa + 3\alpha t)$ .  $S_2$  and  $S_3$  survive:  $r_{\max} = 3$ .
- (iii) **Class M** ( $\Delta \neq 0$ ).  $Q_L$  is irreducible over  $k(c)$ ;  $\sqrt{Q_L}$  is irrational;  $S_r \neq 0$  for all  $r \geq 4$  generically:  $r_{\max} = \infty$ .

For  $r \geq 4$ ,

$$S_r = \Delta \cdot R_r, \quad R_r \in \mathbb{Q}(\alpha, \kappa, S_4). \quad (3.3)$$

*Proof.* The classification follows from Theorem 3.1 and the Gaussian decomposition (Corollary 3.3).

*Case  $\Delta = 0$ .* Then  $Q_L = (2\kappa + 3\alpha t)^2$  is a perfect square, so  $\sqrt{Q_L} = 2\kappa + 3\alpha t$  is polynomial and  $H(t) = t^2(2\kappa + 3\alpha t)$  terminates. If  $\alpha = 0$ :  $H(t) = 2\kappa t^2$ , giving  $r_{\max} = 2$ . If  $\alpha \neq 0$ :  $H(t) = 2\kappa t^2 + 3\alpha t^3$ , giving  $r_{\max} = 3$ .

Case  $\Delta \neq 0$ . The discriminant of  $Q_L$  as a polynomial in  $t$  is

$$(12\kappa\alpha)^2 - 4 \cdot 4\kappa^2 \cdot (9\alpha^2 + 16\kappa S_4) = -32\kappa^2\Delta \neq 0,$$

so  $Q_L$  is irreducible over  $k(c)$ . The Taylor expansion of  $\sqrt{Q_L}$  around  $t = 0$  has nonzero coefficients at all orders generically, hence  $r_{\max} = \infty$ .

*Universal factorization.* The binomial expansion of  $\sqrt{Q_L}$  around the Gaussian envelope  $(2\kappa + 3\alpha t)$  is

$$\sqrt{Q_L} = (2\kappa + 3\alpha t) \sqrt{1 + \frac{2\Delta t^2}{(2\kappa + 3\alpha t)^2}}.$$

Expanding  $\sqrt{1+u}$  as  $1 + \frac{1}{2}u - \frac{1}{8}u^2 + \dots$  with  $u = 2\Delta t^2 / (2\kappa + 3\alpha t)^2$ , every coefficient of  $t^n$  for  $n \geq 2$  in  $\sqrt{Q_L} - (2\kappa + 3\alpha t)$  carries a factor of  $\Delta$ . Since  $rS_r = [t^{r-2}] \sqrt{Q_L}$  for  $r \geq 2$ , and the Gaussian part  $2\kappa + 3\alpha t$  contributes only at  $r \leq 3$ , the factorization  $S_r = \Delta \cdot R_r$  for  $r \geq 4$  follows.  $\square$

*Remark 3.8* (The contact class). The partition  $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}$  of chirally Koszul algebras includes a fourth class  $\mathbf{C}$  (contact,  $r_{\max} = 4$ ), realized by the  $\beta\gamma$  system. This class escapes the single-line dichotomy by *stratum separation*: the quartic contact invariant lives on a charged stratum of  $\text{Def}_{\text{cyc}}^{\text{mod}}$  where  $\kappa = 0$ , and the shadow metric requires  $\kappa \neq 0$ . On the charged stratum, the self-bracket maps to a zero-dimensional target by rank-one rigidity, so the quartic pump does not activate and the tower terminates at  $r = 4$ . The algebraic shadow depth  $d_{\text{alg}} \in \{0, 1, 2, \infty\}$  exhausts the four classes:  $\mathbf{G}$  ( $d_{\text{alg}} = 0$ ),  $\mathbf{L}$  ( $d_{\text{alg}} = 1$ ),  $\mathbf{C}$  ( $d_{\text{alg}} = 2$ ),  $\mathbf{M}$  ( $d_{\text{alg}} = \infty$ ).

**3.6. The shadow growth rate.** For class  $\mathbf{M}$  algebras, the shadow obstruction tower is infinite but governed by a single analytic invariant: the growth rate of the coefficients  $S_r$ .

**Definition 3.9** (Shadow growth rate). Let  $L$  be a primary line of class  $\mathbf{M}$  ( $\Delta \neq 0$ ), and write  $Q_L(t) = q_0 + q_1 t + q_2 t^2$  for the shadow metric. The *shadow growth rate* of  $L$  is

$$\rho_L := \sqrt{\frac{q_2}{q_0}} = \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

By Vieta's theorem, the zeros  $t_0, \bar{t}_0$  of  $Q_L$  satisfy  $|t_0|^2 = q_0/q_2$ , so  $\rho_L = 1/|t_0|$ : the reciprocal of the branch-point modulus. For classes  $\mathbf{G}$  and  $\mathbf{L}$ , set  $\rho := 0$  by convention.

**Theorem 3.10** (Shadow asymptotics). Let  $L$  be a primary line of class  $\mathbf{M}$  with shadow growth rate  $\rho_L$ .

(i) Asymptotic formula.

$$S_r \sim A(\mathcal{A}) \rho_L^r r^{-5/2} \cos(r\theta + \varphi) \quad (r \rightarrow \infty),$$

where  $\theta := -\arg(t_0)$  is the oscillation phase,  $A(\mathcal{A}) > 0$ , and  $\varphi$  is computable from the branch-point residue of  $\sqrt{Q_L}$ . The exponent  $-5/2$  arises from  $S_r = [t^{r-2}] \sqrt{Q_L}/r$  combined with the  $n^{-3/2}$  transfer law for square-root singularities.

(ii) Unit circle criterion.  $\rho_L = 1$  if and only if  $q_0 = q_2$ , i.e.,

$$(2\kappa - 3\alpha)(2\kappa + 3\alpha) = 2\Delta.$$

Equivalently, the branch points of  $Q_L$  lie on the unit circle  $|t| = 1$ .

(iii) Convergence partition.  $\rho_L < 1$ : the shadow obstruction tower converges absolutely.  $\rho_L > 1$ : the shadow obstruction tower diverges, but the algebraic function  $\sqrt{Q_L(t)}$  extends analytically past the convergence radius.  $\rho_L = 1$ : marginal case (the critical surface).

*Proof.* The function  $\sqrt{Q_L(t)}$  has algebraic branch-cut singularities of order  $\frac{1}{2}$  at each simple zero  $t_0$  of  $Q_L$ . Since  $Q_L$  is quadratic with nonzero discriminant ( $\Delta \neq 0$  implies  $\text{disc}(Q_L) = -32\kappa^2\Delta \neq 0$ ), the two zeros  $t_0, \bar{t}_0$  are distinct with  $|t_0| = |\bar{t}_0| = \sqrt{q_0/q_2} = 1/\rho_L$ .

The transfer theorem for algebraic singularities [9, Theorem VI.1] gives  $[t^n]\sqrt{Q_L} \sim C_0 t_0^{-n} n^{-3/2}$  for each dominant singularity  $t_0$  at distance  $R = 1/\rho_L$  from the origin. Both branch points contribute (equal modulus), producing the cosine interference term from  $2 \operatorname{Re}(C_0 t_0^{-(r-2)})$ . Since  $S_r = [t^{r-2}]\sqrt{Q_L}/r$  (Corollary 3.4), the extra factor  $1/r$  gives the stated  $r^{-5/2}$  exponent.

For (ii):  $\rho_L = 1$  iff  $q_2/q_0 = 1$ . Writing  $q_0 = 4\kappa^2$  and  $q_2 = 9\alpha^2 + 2\Delta$ , the condition  $q_0 = q_2$  factors as  $4\kappa^2 - 9\alpha^2 = 2\Delta$ . Part (iii) is immediate from the definition of the convergence radius  $R = 1/\rho_L$ .  $\square$

### 3.7. The depth decomposition.

**Theorem 3.11** (Depth decomposition). *The shadow depth of a chirally Koszul algebra  $\mathcal{A}$  decomposes as*

$$d(\mathcal{A}) = 1 + d_{\text{arith}}(\mathcal{A}) + d_{\text{alg}}(\mathcal{A}),$$

where:

- (i)  $d_{\text{arith}}$  counts the independent holomorphic Hecke eigenforms in the spectral decomposition of the completed shadow partition function  $\widehat{Z}_{\mathcal{A}}^c$  on  $\mathcal{M}_{1,1}$ ;
- (ii)  $d_{\text{alg}} \in \{0, 1, 2, \infty\}$  is the algebraic depth, determined by the single-line dichotomy (Theorem 3.7) and stratum separation (Remark 3.8):

$$d_{\text{alg}} = \begin{cases} 0 & \text{class } \mathbf{G}, \\ 1 & \text{class } \mathbf{L}, \\ 2 & \text{class } \mathbf{C}, \\ \infty & \text{class } \mathbf{M}; \end{cases}$$

- (iii) depths  $d \geq 5$  are purely arithmetic: they arise from  $d_{\text{arith}}$ , not from the  $A_\infty$  structure.

*Proof.* The algebraic depth  $d_{\text{alg}}$  is determined by the structure of the cyclic deformation complex and the MC recursion: the single-line dichotomy (Theorem 3.7) gives  $d_{\text{alg}} \in \{0, 1, \infty\}$  on each primary line, and stratum separation (Remark 3.8) provides  $d_{\text{alg}} = 2$  for the contact class. Since  $d_{\text{alg}} \leq 2$  for finite cases ( $\mathbf{G}/\mathbf{L}/\mathbf{C}$ ) and the base term is 1 (the curvature  $\kappa$  at degree 2), the maximum finite depth without arithmetic contributions is  $1 + 0 + 2 = 3$ ; hence all depths  $d \geq 5$  require  $d_{\text{arith}} \geq 2$ .

The arithmetic depth counts independent holomorphic Hecke eigenforms by decomposing  $\widehat{Z}_{\mathcal{A}}^c$  via the Roelcke–Selberg spectral decomposition on  $\mathcal{M}_{1,1}$ . Each holomorphic eigenform of weight  $k$  contributes a functional-equation centre at  $\operatorname{Re}(s) = k/2$ ; Maass cusp-form projections contribute no new constituent centres.  $\square$

**Example 3.12** (Standard families). The shadow metric data for the standard families:

| Family                            | $\kappa$                                      | $\alpha$ | $S_4$                 | $\Delta$           | Class    | $\rho$                                      |
|-----------------------------------|-----------------------------------------------|----------|-----------------------|--------------------|----------|---------------------------------------------|
| Heisenberg $\mathcal{H}_k$        | $k$                                           | 0        | 0                     | 0                  | <b>G</b> | 0                                           |
| Affine $\widehat{\mathfrak{g}}_k$ | $\frac{\dim \mathfrak{g}(k+h^\vee)}{2h^\vee}$ | $\neq 0$ | 0                     | 0                  | <b>L</b> | 0                                           |
| $\beta\gamma$ (weight $\lambda$ ) | $6\lambda^2 - 6\lambda + 1$                   | $0^*$    | $0^*$                 | $0^*$              | <b>C</b> | 0                                           |
| Virasoro $\operatorname{Vir}_c$   | $c/2$                                         | 2        | $\frac{10}{c(5c+22)}$ | $\frac{40}{5c+22}$ | <b>M</b> | $\frac{1}{c} \sqrt{\frac{180c+872}{5c+22}}$ |

\* The table reports the weight-changing primary line, where  $\kappa = S_3 = S_4 = 0$ . The quartic contact invariant  $Q^{\text{contact}} \neq 0$  lives on this charged stratum; the shadow metric requires  $\kappa \neq 0$  and does not

apply. The class **C** designation ( $r_{\max} = 4$ ) comes from rank-one rigidity, not the single-line dichotomy. The T-line (stress tensor) carries the Virasoro shadow data  $(\kappa, \alpha, S_4) = (c/2, 2, 10/(c(5c+22)))$  with  $c = 2(6\lambda^2 - 6\lambda + 1)$  and is class **M**.

For the Virasoro algebra, the shadow metric evaluates to

$$Q_{\text{Vir}}(t) = c^2 + 12c t + \frac{180c + 872}{5c + 22} t^2,$$

with Gaussian decomposition  $Q_{\text{Vir}} = (c + 6t)^2 + \frac{80}{5c+22} t^2$ . The critical central charge  $c^*$  where  $\rho = 1$  is the unique positive real root of

$$5c^3 + 22c^2 - 180c - 872 = 0,$$

approximately  $c^* \approx 6.12$ . For  $c > c^*$  the shadow tower converges; for  $c < c^*$  it diverges. At the Koszul self-dual point  $c = 13$ :  $\rho(\text{Vir}_{13}) \approx 0.467$ .

*Remark 3.13* (Large- $c$  asymptotic of  $S_4$ ). The Virasoro quartic shadow

$$S_4(\text{Vir}_c) = \frac{10}{c(5c + 22)}$$

has large- $c$  asymptotic

$$S_4(\text{Vir}_c) = \frac{10}{5c^2} \left( 1 - \frac{22}{5c} + O(c^{-2}) \right) = \frac{2}{c^2} + O(c^{-3}),$$

not  $2/(5c^2)$ ; the factor of 5 in the denominator of  $10/[c(5c + 22)]$  is absorbed when expanding  $1/(5c + 22)$  at large  $c$ . Pairing this with  $\kappa = c/2$  gives the critical discriminant  $\Delta = 8\kappa S_4 = 40/(5c + 22) \rightarrow 8/c$  as  $c \rightarrow \infty$ , confirming class **M** mixing persists to leading order.

The identity  $S_2(\text{Vir}_c) = \kappa = c/2$  is convention-independent. The occasional error  $S_2 = c/12$  conflates the shadow invariant with the Vol II lambda-bracket divided-power coefficient  $c/12 = (c/2)/3!$ , which is a different normalisation; the Vol I–II bridge is  $S_2^{\text{Vol I}} = 6 S_2^{\lambda}$ .

The Riccati seed on  $\text{Vir}_c$  is  $(r_o, S_{r_o}) = (3, 2)$ :  $S_3(\text{Vir}_c) = 2$  is the  $c$ -independent cubic shadow (Lorgat, *The Virasoro spectral R-matrix*), and the shadow-exponential base  $C_A = 6$  on the non-logarithmic class **M** locus factorises as  $6 = r_o \cdot S_{r_o} = 3 \cdot 2$ .

*Remark 3.14* (Spectral curve). The algebraic relation  $H^2 = t^4 Q_L(t)$  defines, for each primary line  $L$ , a spectral curve  $\Sigma_L := \{H^2 = t^4 Q_L(t)\} \subset \mathbb{A}_{t,H}^2$ . Since  $Q_L$  is quadratic in  $t$ ,  $\Sigma_L$  has arithmetic genus 0; the projection  $(t, H) \mapsto t$  is a double cover ramified at the zeros of  $Q_L$ . The degeneration loci in parameter space are controlled by the critical discriminant:  $\Sigma_L$  acquires a node precisely when  $\Delta = 0$ . The shadow growth rate  $\rho_L$  is the reciprocal of the distance from  $t = 0$  to the nearest ramification point.

#### 4. MAIN RESULTS

**Theorem 4.1** (Main theorem: algebraicity, classification, and genus-2 formula). *Let  $\mathcal{A}$  be a chirally Koszul vertex algebra on a smooth projective curve  $X$ . Let  $\kappa = \kappa(\mathcal{A})$  be the modular characteristic, and let  $\alpha, S_4$  be the cubic and quartic shadow invariants. Define the shadow metric*

$$Q(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4) t^2$$

and the critical discriminant  $\Delta = 8\kappa S_4$ .

(i) **Algebraicity.** *The weighted shadow generating function  $H(t) = \sum_{r \geq 2} r S_r t^r$  satisfies*

$$H(t)^2 = t^4 Q(t).$$

*Every shadow coefficient  $S_r$  for  $r \geq 5$  is determined by  $(\kappa, \alpha, S_4)$  via  $S_r = \frac{1}{r} [t^{r-2}] \sqrt{Q(t)}$ .*

- (ii) **Shadow connection.** The shadow metric defines a logarithmic connection  $\nabla^{\text{sh}} = d - Q'/(2Q) dt$  with residue  $\frac{1}{2}$  at each zero of  $Q$ . Monodromy  $= -1$  (Koszul sign).
- (iii) **Four-class partition.** Write  $Q = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ . The shadow depth  $r_{\max}$  lies in  $\{2, 3, 4, \infty\}$ :
- Class **G**:  $\Delta = 0, \alpha = 0, r_{\max} = 2$ .
  - Class **L**:  $\Delta = 0, \alpha \neq 0, r_{\max} = 3$ .
  - Class **C**:  $\Delta = 0$  on the primary line; quartic contact on a charged stratum,  $r_{\max} = 4$ .
  - Class **M**:  $\Delta \neq 0, r_{\max} = \infty$ .
- (iv) **Universal factorization.** For  $r \geq 4$ :  $S_r = \Delta \cdot R_r$ , where  $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$  is independent of the algebra.
- (v) **Planted-forest formula (conditional).** Under the determinant-line sign and residue-pushforward transport package, the genus-2 planted-forest correction on a primary line is predicted to be  $\delta_{\text{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$ . It vanishes for class **G** (where  $S_3 = 0$ ).

*Proof.* Parts (i)–(ii) are proved in Theorem 3.1 and Proposition 3.5, respectively. Part (iii) follows from Theorem 3.7. Part (iv) is the universal factorization in Theorem 3.7, equation (3.3). Part (v) is conditional on the orientation and pushforward package named in the statement.  $\square$

*Remark 4.2* (Three independent depths). The inequalities  $p_{\max} \neq k_{\max} \neq r_{\max}$  separate OPE pole order, collision depth, and shadow depth. The shadow classification is the stratification by the third invariant; the operator-order trichotomy is the stratification by the first two.

## 5. THE FOUR SHADOW CLASSES

The single-line Riccati dichotomy partitions noncritical one-dimensional primary lines into the three rank-one outcomes G, L, and M. The fourth standard shadow class, C, is produced by charged-stratum separation in the  $bc$  and  $\beta\gamma$  families. The rank-one outcomes are determined by two invariants of the shadow metric  $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ : the cubic coefficient  $\alpha = S_3$  and the critical discriminant  $\Delta = 8\kappa S_4$ .

**Definition 5.1** (Shadow classes). Let  $\mathcal{A}$  be a chirally Koszul algebra with shadow metric  $Q_L$  on a one-dimensional primary line  $L$ . The *shadow class* of  $\mathcal{A}$  along  $L$  is:

| Class          | $\alpha$ | $\Delta$  | $r_{\max}$ | mechanism               |
|----------------|----------|-----------|------------|-------------------------|
| $G$ (Gaussian) | 0        | 0         | 2          | constant metric         |
| $L$ (Lie/tree) | $\neq 0$ | 0         | 3          | square metric           |
| $C$ (Contact)  | 0        | line data | 4          | charged contact stratum |
| $M$ (Mixed)    | $\neq 0$ | $\neq 0$  | $\infty$   | irreducible quadratic   |

Here  $r_{\max}$  is the *shadow depth*: the largest degree  $r$  for which the shadow coefficient  $S_r \neq 0$ . Class C is the finite contact stratum of the standard  $bc_\lambda$  and  $\beta\gamma_\lambda$  families, where  $S_3 = 0$ ,  $S_4 = -5/12$ , and  $S_r = 0$  for  $r \geq 5$  on the standard conformal-weight channel.

The classification is exhaustive: every entry in Table 1 falls into exactly one class, and every class is realized by at least one standard family. The mechanism is algebraic: the shadow generating function  $H(t) = t^2 \sqrt{Q_L(t)}$  is algebraic of degree 2 on each rank-one primary line, and its Taylor coefficients  $S_r$  are computed from the convolution recursion  $H(t)^2 = t^4 Q_L(t)$ , where  $H(t) = \sum_{r \geq 2} S_r t^r$  and  $S_r = a_{r-2}/r$  with  $a_n$  the Taylor coefficients of  $\sqrt{Q_L}$ . When  $Q_L$  is a perfect square ( $\Delta = 0$ ), the square root terminates; when  $Q_L$  is irreducible ( $\Delta \neq 0$ ), the Taylor expansion is infinite.

### 5.1. Class G: Gaussian ( $r_{\max} = 2$ ).

**Definition 5.2.** A chirally Koszul algebra  $\mathcal{A}$  is *Gaussian* if  $S_r(\mathcal{A}) = 0$  for all  $r \geq 3$ . Equivalently,  $\alpha = 0$  and  $\Delta = 0$ , so the shadow metric  $Q_L(t) = 4\kappa^2$  is constant and the shadow obstruction tower terminates at the scalar curvature.

**Examples.** Heisenberg  $\mathcal{H}_k$  with  $\kappa = k$ ; lattice VOAs  $V_\Lambda$  with  $\kappa = \text{rank}(\Lambda)$ ; free fermion  $\psi$  with  $\kappa = 1/4$ .

**Characterization.** Class G algebras are freely generated as chiral algebras. Their bar complex is quadratic (the differential has no higher-order terms), and the OPE on each primary line is abelian. The full obstruction is captured by a single scalar:

$$\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda,$$

where  $\eta$  is the cyclic direction and  $\Lambda \in H^*(\bar{\mathcal{M}}_g)$  is the tautological class. All genus- $g$  data is controlled by  $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$  (uniform-weight).

### 5.2. Class L: Lie/tree ( $r_{\max} = 3$ ).

**Definition 5.3.** A chirally Koszul algebra  $\mathcal{A}$  is *Lie/tree* if  $\alpha \neq 0$  and  $\Delta = 0$ , so that  $S_3 \neq 0$  but  $S_r = 0$  for all  $r \geq 4$ . The shadow metric is a perfect square:  $Q_L(t) = (2\kappa + 3\alpha t)^2$ .

**Examples.** Affine Kac–Moody  $V_k(\mathfrak{g})$  for every simple Lie algebra  $\mathfrak{g}$  and every non-critical level  $k \neq -h^\vee$ : type  $A$  ( $\mathfrak{sl}_N$ ), types  $B, C, D$  (orthogonal and symplectic), exceptional types  $E_6, E_7, E_8, G_2, F_4$ . For all of these:

$$\kappa(V_k(\mathfrak{g})) = \frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}.$$

**Characterization.** The nonzero cubic shadow  $S_3 = \alpha$  arises from the Lie bracket (structure constants  $f_{abc}$ ), while the Jacobi identity forces the quartic shadow  $S_4 = 0$  on the primary line. In trace form,

$$r_{\text{tr}}^{\text{KM}}(z) = k \Omega_{\text{tr}}/z.$$

The KZ-normalized connection uses

$$r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k + h^\vee)z), \quad \Omega_{\text{KZ}} = 2h^\vee \Omega_{\text{tr}}.$$

These are conventionally related normalizations with different  $k$ -dependence. The tree-level graph sum terminates at trivalent vertices.

### 5.3. Class C: Contact ( $r_{\max} = 4$ ).

**Definition 5.4.** A chirally Koszul algebra  $\mathcal{A}$  is *contact* if  $S_3 = 0$ ,  $S_4 \neq 0$ , and  $S_r = 0$  for  $r \geq 5$  on the standard contact channel. In the verified standard families this is the conformal-weight channel of  $bc_\lambda$  and  $\beta\gamma_\lambda$ , where  $S_4 = -5/12$ .

**Mechanism.** Class C is a finite contact phenomenon external to the rank-one Riccati dichotomy. The weight-changing line is abelian, while the standard conformal-weight channel carries the quartic contact coefficient. Rank-one rigidity kills the next charged target, so the tower stops at degree 4.

**Examples.** The  $\beta\gamma$  system at conformal weight  $\lambda$ , with

$$\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1 = \frac{1}{2} c(\beta\gamma_\lambda),$$

and its Koszul dual, the  $bc$  system at the same weight. At the standard weight  $\lambda = 0$  or  $\lambda = 1$ :  $c = 2$  and  $\kappa = 1$ .

**5.4. Class FF: Feigin–Frenkel locus ( $\kappa = 0$ ).** The four classes  $G, L, C, M$  stratify the non-critical locus  $\kappa(\mathcal{A}) \neq 0$ . A fifth class appears at the boundary  $\kappa = 0$ : the *Feigin–Frenkel locus* where the Sugawara topologisation fails.

**Definition 5.5** (Class FF). A chirally Koszul algebra  $\mathcal{A}$  belongs to class FF if it lies at a critical Feigin–Frenkel centre or coset locus where the Sugawara topologisation degenerates. The vanishing  $\kappa(\mathcal{A}) = 0$  is a necessary scalar signal in the affine examples; the class itself is controlled by the Feigin–Frenkel coset data  $\text{coset}_{FF}(\mathcal{A})$ .

**Examples.** The critical-level affine algebra  $V_{-h^\vee}(\mathfrak{g})$  for every simple  $\mathfrak{g}$  (so  $V_{-2}(\mathfrak{sl}_2), V_{-3}(\mathfrak{sl}_3), \dots$ ); its commutant lies in the Feigin–Frenkel centre  $\mathfrak{z}(\widehat{\mathfrak{g}})$ , polynomial in infinitely many Sugawara-like generators. The Koszul dual of class L under the involution  $k \mapsto -k - 2h^\vee$  passes through class FF at the fixed point  $k = -h^\vee$ . Critical  $\mathcal{W}_N$  (at the Feigin–Frenkel image of the affine critical level) is also class FF.

**Characterization.** At the Feigin–Frenkel critical locus the conformal vector degenerates: the Sugawara construction  $T = (k + h^\vee)^{-1} : J^a J^a$  has a pole, and topologisation  $\text{SC}^{\text{ch}, \text{top}} \rightarrow E_3^{\text{top}}$  fails. The coset slot of the canonical fingerprint switches from  $(\text{Com}(\mathcal{H}_\kappa, \mathcal{A}), \mathcal{A}/\text{Com})$  to the Feigin–Frenkel coset  $\text{coset}_{FF}(\mathcal{A}) \subset \mathfrak{z}(\widehat{\mathfrak{g}})$ . Class FF is the degeneration locus for the ambient shadow invariants. On this locus the controlling datum is the Feigin–Frenkel coset data instead of  $(\alpha, \Delta)$ .

**Remark 5.6** (FF as a fifth companion class). The four classes  $G, L, C, M$  are determined by the shadow metric  $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$ , which degenerates at  $\kappa = 0$ : the leading constant term vanishes, the Riccati algebraicity argument loses its pivot, and the tower of  $\Theta_{\mathcal{A}}^{\leq r}$  must be read from a different source. Together with  $G, L, C, M$ , this companion locus gives the five-class fingerprint stratification used in the standard landscape.

**5.5. Class M: Mixed ( $r_{\max} = \infty$ ).**

**Definition 5.7.** A chirally Koszul algebra  $\mathcal{A}$  is *mixed* if  $\alpha \neq 0$  and  $\Delta \neq 0$ , so the shadow metric  $Q_L$  is an irreducible quadratic in  $t$  and the shadow obstruction tower does not terminate.

**Examples.** Virasoro  $\text{Vir}_c$  with  $\kappa = c/2$ ;  $\mathcal{W}$ -algebras  $\mathcal{W}_N$  with  $\kappa = (H_N - 1)c$  where  $H_N = \sum_{j=1}^N 1/j$ ; Drinfeld–Sokolov reductions of affine Kac–Moody algebras.

**Characterization.** The shadow growth rate

$$\rho(\mathcal{A}) = \frac{\sqrt{9\alpha^2 + 16\kappa S_4}}{2|\kappa|}$$

controls the asymptotic behaviour of  $S_r$ : the coefficients satisfy  $S_r \sim C(\mathcal{A}) \rho(\mathcal{A})^r r^{-5/2} \cos(r\theta + \phi)$  for a branch-point argument  $\theta$  determined by the shadow metric. The critical central charge  $c^*$  where  $\rho(\text{Vir}_c) = 1$  is the unique positive real root of the cubic

$$5c^3 + 22c^2 - 180c - 872 = 0, \quad c^* \approx 6.125. \quad (5.1)$$

For  $c > c^*$  the tower converges ( $\rho < 1$ ), and for  $c < c^*$  it diverges ( $\rho > 1$ ). At the self-dual point  $c = 13$ :  $\rho(\text{Vir}_{13}) \approx 0.467$ .

The Virasoro shadow obstruction tower has explicitly computed coefficients through degree 6:

$$\begin{aligned} S_2 &= \frac{c}{2}, & S_3 &= 2, & S_4 &= \frac{10}{c(5c + 22)}, \\ S_5 &= \frac{-48}{c^2(5c + 22)}, & S_6 &= \frac{4(240c + 1031)}{c^3(5c + 22)^2}. \end{aligned}$$

The identity  $S_3 = 2$  is independent of  $c$  by direct residue extraction from the Virasoro OPE.

*Remark 5.8* (Depth and Koszulness). Shadow depth  $r_{\max}$  does *not* characterize chiral Koszulness. All four classes (Gaussian,  $r_{\max} = 2$ ; Lie,  $r_{\max} = 3$ ; contact,  $r_{\max} = 4$ ; mixed,  $r_{\max} = \infty$ ) consist of chirally Koszul algebras. Shadow depth classifies *complexity* within the Koszul world: the operadic complexity conjecture identifies  $r_{\max}(\mathcal{A})$  with the  $A_\infty$ -depth and  $L_\infty$ -formality level.

*Remark 5.9* (Total depth decomposition). The total shadow depth decomposes as  $d(\mathcal{A}) = 1 + d_{\text{arith}} + d_{\text{alg}}$ , where the algebraic depth  $d_{\text{alg}} \in \{0, 1, 2, \infty\}$  and depths  $\geq 5$  are purely arithmetic (cusp forms contribute beyond the algebraic engine).

**5.6. The quadrichotomy with logarithmic and Wakimoto companions.** The four classes  $G, L, C, M$  together with the Feigin–Frenkel locus FF form the fingerprint stratification of the non-logarithmic standard landscape. Two further labels occur on enlarged surfaces: logarithmic  $\mathcal{W}(p)$  behaviour and Wakimoto free-field realisations. They are companion overlays that may overlap the five strata.

**Definition 5.10** (Class  $M^*$ : logarithmic companion). A chirally Koszul algebra  $\mathcal{A}$  carries the companion label  $M^*$  if it is logarithmic in the sense of Gurarie–Flohr [2, 5], i.e. the Zhu algebra  $\mathcal{A}(\mathcal{A})$  is finite-dimensional but the Hochschild Massey products  $\langle \Omega, \Omega, \dots, \Omega \rangle$  carry amplitude on arbitrarily high tensor powers of the vacuum.

**Examples.** Triplet algebras  $\mathcal{W}(p)$  for  $p \geq 2$ ; the symplectic-fermion pair at central charge  $c = -2$ ; logarithmic parafermion cosets.

**Characterization.** Class  $M^*$  algebras satisfy  $\kappa \neq 0$  and share the mixed-shadow expansion  $\alpha \neq 0$ ,  $\Delta \neq 0$  of class M, but are distinguished by logarithmic amplitudes: Zhu-finite yet Massey-unbounded. The non-logarithmic shadow analysis applies on the formal-weight line; logarithmic corrections leave the algebraic invariants  $\alpha = S_3$  and  $\Delta = 8\kappa S_4$ , which remain finite and computable, but they preclude tempering of the shadow tower.

**Definition 5.11** (Class  $\mathcal{W}$ : Wakimoto stratum). A chirally Koszul algebra  $\mathcal{A}$  belongs to class  $\mathcal{W}$  if it arises as the screening-kernel of a rank- $(N - 1)$  Heisenberg through Feigin–Frenkel–Wakimoto realisation,  $\mathcal{A} = \bigcap_i \ker Q_{\alpha_i} \subset \mathcal{H}^{\oplus(N-1)}$ .

**Examples.** Principal  $\mathcal{W}_N$  as a Wakimoto kernel inside  $\mathcal{H}^{\oplus(N-1)}$  [3, 4]; affine  $\widehat{\mathfrak{sl}}_2$  through the  $\beta\gamma\phi$  Wakimoto realisation; admissible affine algebras through parabolic screenings [1, 12].

**Characterization.** Class  $\mathcal{W}$  can overlap  $G, L, C, M$ : it is a presentational overlay. The ambient Heisenberg is class  $G$ , while the screening kernel may land in class L or class M after the BRST and screening-obstruction data are imposed. The label records the free-field presentation; Koszulity and Drinfeld coproduct descent require separate screening-kernel input.

*Remark 5.12* (Fingerprint strata and overlays). The four shadow-depth classes  $G, L, C, M$  stratify the non-logarithmic non-critical locus by the algebraic invariants  $(\alpha, \Delta)$  of the shadow metric. Class FF marks the  $\kappa = 0$  critical boundary. Class  $M^*$  records logarithmic amplitude behaviour orthogonal to  $(\alpha, \Delta)$ : a Zhu-finite class-M algebra is class-M, class- $M^*$ , or both, depending on Massey boundedness [2, 5]. Class  $\mathcal{W}$  records a presentation overlay: a class-L or class-M algebra may simultaneously be class- $\mathcal{W}$  when realised as a screening kernel. The stratification is five-class:  $G, L, C, M, FF$ ;  $M^*$  and  $\mathcal{W}$  are companion labels.

**5.7. Leading T-line exponential base.** The leading large-central-charge coefficients on the Virasoro stress-tensor line have a sharp exponential base. This is a T-line statement, separate from exact finite- $c$  radius questions for the full shadow series of an arbitrary class-M algebra.

**Proposition 5.13** (Leading T-line exponential base  $C_{\mathcal{A}}^T = 6$ ). *Let  $\mathcal{A}$  be a non-logarithmic class-M standard family with a generic Virasoro stress-tensor line, and suppose the T-line dominance hypotheses hold. Let  $A_r^T$  denote the leading large- $c$  coefficient of the T-line shadow coefficient  $S_r^T$ . Then*

$$C_{\mathcal{A}}^T := \limsup_{r \rightarrow \infty} |A_{r+1}^T / A_r^T| = 6.$$

*For the Virasoro line the leading generating function has the logarithmic singularity*

$$\Sigma_{\text{Vir}}^{T,\text{lead}}(z) = \frac{4z^3}{9} - \frac{z^2}{9} + \frac{z}{27} - \frac{\log(1+6z)}{162}.$$

*Proof.* The T-line large- $c$  recursion is the Virasoro Riccati recursion with seed  $S_3 = 2$ . Solving the recursion gives the displayed logarithmic term, whose nearest singularity is at  $z = -1/6$ . The root test gives the exponential base 6 for the leading coefficients  $A_r^T$ .  $\square$

For  $\mathcal{W}_N$  the corresponding Vol II tempering exponent is a separate invariant. Under the harmonic tempering law it is

$$\beta_{\mathcal{W}_N} = 12(H_N - 1),$$

so  $\beta_{\mathcal{W}_3} = 10$  and  $\beta_{\mathcal{W}_4} = 13$ . It is not the value given by the polynomial interpolant  $(N+1)(N+2)/2$ .

**5.8. Picard–Fuchs spectral-curve presentation.** The shadow metric  $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$  is a rank-one spectral datum; its square root  $\sqrt{Q_L}$  is the Seiberg–Witten-type differential of a spectral cover. On non-logarithmic class M, the shadow generating function  $\Sigma_{\mathcal{A}}(z)$  satisfies a Picard–Fuchs equation whose monodromy encodes the Borel geometry of the shadow tower.

**Proposition 5.14** (Picard–Fuchs equation for the leading T-line). *Let  $\mathcal{A}$  be class M with a generic Virasoro T-line and suppose the leading T-line dominance hypotheses of Proposition 5.13 hold. The leading T-line shadow series  $\Sigma_{\mathcal{A}}^{T,\text{lead}}(z)$  satisfies a second-order Picard–Fuchs-type equation*

$$(1+6z)^2 (\Sigma_{\mathcal{A}}^{T,\text{lead}})'' + P_1(z) (\Sigma_{\mathcal{A}}^{T,\text{lead}})' + P_0(z) \Sigma_{\mathcal{A}}^{T,\text{lead}} = R_{\mathcal{A}}(z),$$

*with  $P_0, P_1, R_{\mathcal{A}}$  polynomials of degree at most 2 in  $z$  determined by the leading T-line Riccati data. The regular-singular point  $z = -1/6$  is the leading T-line branch point.*

*Proof.* Differentiate the logarithmic expression in Proposition 5.13. Clearing the denominator  $(1+6z)^2$  gives the displayed inhomogeneous second-order equation with polynomial coefficients. The only singularity introduced by the logarithmic term in the leading T-line expression is  $z = -1/6$ .  $\square$

## 6. SHADOW TABLES

The modular characteristic  $\kappa(\mathcal{A})$  is the degree-2 projection of the universal MC element  $\Theta_{\mathcal{A}}$ , and the genus- $g$  free energy is  $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$  for single-generator (uniform-weight) algebras at all genera, and unconditionally at genus 1 for all families. Here  $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$  are the Faber–Pandharipande intersection numbers.

**6.1. Master shadow table.** Table 1 records, for each standard family, the modular characteristic  $\kappa$ , the shadow class, the shadow depth  $r_{\text{max}}$ , the cubic coefficient  $\alpha = S_3$ , the quartic coefficient  $S_4$ , and the critical discriminant  $\Delta = 8\kappa S_4$ .

For  $bc_{\lambda}$  and  $\beta\gamma_{\lambda}$ , the class-C witness is the standard conformal-weight family:  $S_2 = 6\lambda^2 - 6\lambda + 1$ ,  $S_3 = 0$ ,  $S_4 = -5/12$ , and  $S_r = 0$  for  $r \geq 5$ , with the sign of  $\Delta = 8\kappa S_4$  set by the sign of  $\kappa$ .

In the table,  $H_N = 1 + 1/2 + \cdots + 1/N$  denotes the  $N$ -th harmonic number. The Virasoro quartic contact  $S_4 = Q_{\text{Vir}}^{\text{contact}} = 10/[c(5c+22)]$  is the unique quartic shadow on the T-line, proved by direct

TABLE I. Master shadow table: standard landscape

| Algebra $\mathcal{A}$                    | $\kappa(\mathcal{A})$      | Class | $r_{\max}$ | $S_3$    | $S_4$                 | $\Delta$                               |
|------------------------------------------|----------------------------|-------|------------|----------|-----------------------|----------------------------------------|
| <i>Class G (Gaussian)</i>                |                            |       |            |          |                       |                                        |
| Heisenberg $\mathcal{H}_k$               | $k$                        | G     | 2          | 0        | 0                     | 0                                      |
| Lattice $V_\Lambda$                      | $\text{rank}(\Lambda)$     | G     | 2          | 0        | 0                     | 0                                      |
| Free fermion $\psi$                      | $1/4$                      | G     | 2          | 0        | 0                     | 0                                      |
| <i>Class L (Lie/tree)</i>                |                            |       |            |          |                       |                                        |
| $\widehat{\mathfrak{sl}}_2$ at level $k$ | $3(k+2)/4$                 | L     | 3          | $\neq 0$ | 0                     | 0                                      |
| $\widehat{\mathfrak{sl}}_N$ at level $k$ | $(N^2-1)(k+N)/(2N)$        | L     | 3          | $\neq 0$ | 0                     | 0                                      |
| $\widehat{E}_6$ at level $k$             | $13(k+12)/4$               | L     | 3          | $\neq 0$ | 0                     | 0                                      |
| $\widehat{E}_7$ at level $k$             | $133(k+18)/36$             | L     | 3          | $\neq 0$ | 0                     | 0                                      |
| $\widehat{E}_8$ at level $k$             | $62(k+30)/15$              | L     | 3          | $\neq 0$ | 0                     | 0                                      |
| <i>Class C (Contact)</i>                 |                            |       |            |          |                       |                                        |
| $bc$ ghosts (weight $\lambda$ )          | $-(6\lambda^2-6\lambda+1)$ | C     | 4          | 0        | $-\frac{5}{12}$       | $\frac{10}{3}(6\lambda^2-6\lambda+1)$  |
| $\beta\gamma$ (weight $\lambda$ )        | $6\lambda^2-6\lambda+1$    | C     | 4          | 0        | $-\frac{5}{12}$       | $-\frac{10}{3}(6\lambda^2-6\lambda+1)$ |
| <i>Class M (Mixed)</i>                   |                            |       |            |          |                       |                                        |
| $\text{Vir}_c$                           | $c/2$                      | M     | $\infty$   | 2        | $\frac{10}{c(5c+22)}$ | $\frac{40}{5c+22}$                     |
| $\mathcal{W}_3$ at $c$                   | $5c/6$                     | M     | $\infty$   | $\neq 0$ | $\neq 0$              | $\neq 0$                               |
| $\mathcal{W}_4$ at $c$                   | $13c/12$                   | M     | $\infty$   | $\neq 0$ | $\neq 0$              | $\neq 0$                               |
| $\mathcal{W}_5$ at $c$                   | $77c/60$                   | M     | $\infty$   | $\neq 0$ | $\neq 0$              | $\neq 0$                               |
| $\mathcal{W}_N$ at $c$ (general)         | $(H_N-1)c$                 | M     | $\infty$   | $\neq 0$ | $\neq 0$              | $\neq 0$                               |

computation from the Virasoro OPE. The T-line shadow data ( $S_3 = 2, S_4 = 10/[c(5c+22)]$ ) is *universal*: it coincides for  $\mathcal{W}_N$  at all  $N \geq 2$ , since every  $\mathcal{W}_N$  contains the Virasoro subalgebra along the stress-tensor primary line. The additional primary lines (spin-3, spin-4, etc.) carry independent shadow data with separate quartic coefficients.

**6.2. Exceptional and non-simply-laced types.** Table 2 records the affine Kac–Moody data for exceptional and non-simply-laced types. All are class L.

The kappa formula is universal:  $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$ . The central charge sum  $c+c' = 2\dim(\mathfrak{g})$  under the Feigin–Frenkel involution  $k \mapsto -k-2h^\vee$ , and the complementarity  $\kappa+\kappa' = 0$  holds for all affine Kac–Moody algebras. For non-simply-laced types, the dual Coxeter number  $h^\vee$  differs from the Coxeter number  $h$  ( $h^\vee = 3$  for  $B_2$  and  $C_2$ , while  $h = 4$ ). The shadow class is L regardless: the Jacobi identity kills the quartic on every primary line.

*Remark 6.1* ( $B_2/C_2$  isomorphism). The Lie algebras  $B_2 = \mathfrak{so}_3$  and  $C_2 = \mathfrak{sp}_4$  are isomorphic, sharing  $\dim = 10$ ,  $h^\vee = 3$ , and  $h = 4$ . Their Cartan matrices are transposes of each other, but all shadow invariants coincide.

TABLE 2. Shadow data for exceptional and non-simply-laced types

| $\mathfrak{g}$              | $\dim(\mathfrak{g})$ | $h^\vee$ | $\kappa(V_k(\mathfrak{g}))$ | $c(V_k) + c(V_{k'})$ | Class | $r_{\max}$ |
|-----------------------------|----------------------|----------|-----------------------------|----------------------|-------|------------|
| $B_2 \cong \mathfrak{so}_5$ | 10                   | 3        | $5(k+3)/3$                  | 20                   | L     | 3          |
| $C_2 \cong \mathfrak{sp}_4$ | 10                   | 3        | $5(k+3)/3$                  | 20                   | L     | 3          |
| $G_2$                       | 14                   | 4        | $7(k+4)/4$                  | 28                   | L     | 3          |
| $F_4$                       | 52                   | 9        | $26(k+9)/9$                 | 104                  | L     | 3          |
| $E_6$                       | 78                   | 12       | $13(k+12)/4$                | 156                  | L     | 3          |
| $E_7$                       | 133                  | 18       | $133(k+18)/36$              | 266                  | L     | 3          |
| $E_8$                       | 248                  | 30       | $62(k+30)/15$               | 496                  | L     | 3          |

TABLE 3. Complementarity data:  $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$  and  $c(\mathcal{A}) + c(\mathcal{A}^\dagger)$  under Koszul duality

| Family $\mathcal{A}$             | Koszul dual $\mathcal{A}^\dagger$ | $c + c'$               | $\kappa + \kappa'$ | Self-dual point          |
|----------------------------------|-----------------------------------|------------------------|--------------------|--------------------------|
| $\mathcal{H}_k$                  | $\text{Sym}^{\text{ch}}(V^*)$     | $\ddagger$             | o                  | none                     |
| $V_\Lambda$ (lattice)            | $V_{\Lambda^\dagger}$             | $\ddagger$             | o                  | $\Lambda$ self-dual      |
| Free fermion                     | $\text{Sym}^{\text{ch}}(\gamma)$  | o                      | o                  | none                     |
| $\beta\gamma_\lambda$            | $bc_\lambda$                      | o                      | o                  | none                     |
| $V_k(\mathfrak{sl}_2)$           | $V_{-k-4}(\mathfrak{sl}_2)$       | 6                      | o                  | $k = -2$ (critical)      |
| $V_k(\mathfrak{sl}_N)$           | $V_{-k-2N}(\mathfrak{sl}_N)$      | $2(N^2-1)$             | o                  | $k = -N$ (critical)      |
| $V_k(\mathfrak{g})$ (general)    | $V_{-k-2h^\vee}(\mathfrak{g})$    | $2 \dim(\mathfrak{g})$ | o                  | $k = -h^\vee$ (critical) |
| $\text{Vir}_c$                   | $\text{Vir}_{26-c}$               | 26                     | 13                 | $c = 13$                 |
| $\mathcal{W}_3$ at $c$           | $\mathcal{W}_3$ at $100-c$        | 100                    | $250/3$            | $c = 50$                 |
| $\mathcal{W}_4$ at $c$           | $\mathcal{W}_4$ at $246-c$        | 246                    | $533/2$            | $c = 123$                |
| $\mathcal{W}_5$ at $c$           | $\mathcal{W}_5$ at $488-c$        | 488                    | $9394/15$          | $c = 244$                |
| $\mathcal{W}_N$ at $c$ (general) | $\mathcal{W}_N$ at $K_N-c$        | $K_N$                  | $(H_N-1) K_N$      | $c = K_N/2$              |

6.3. **Complementarity table.**  $\ddagger$ The Koszul dual  $\mathcal{H}_k^\dagger = (\text{Sym}^{\text{ch}}(V^*[1]), m_o = -k\omega)$  has  $\kappa(\mathcal{H}_k^\dagger) = -k = \kappa(\mathcal{H}_{-k})$ ; its OPE structure differs from that of  $\mathcal{H}_{-k}$ . The central charge sum has no single-parameter-family meaning.

Here  $K_N = 2(N-1)(2N^2+2N+1)$  is the Koszul conductor for  $\mathcal{W}_N = \mathcal{W}(\mathfrak{sl}_N)$  (the sum  $c + c'$  under the Feigin–Frenkel involution  $k \mapsto -k - 2N$  on the underlying  $\mathfrak{sl}_N$ ). The key structural distinction: for all affine Kac–Moody algebras,  $\kappa + \kappa' = 0$  (anti-symmetry under the Feigin–Frenkel involution); for  $\mathcal{W}$ -algebras,  $\kappa + \kappa' = (H_N - 1) \cdot K_N \neq 0$ . The Virasoro ( $N = 2$ ) gives  $\kappa + \kappa' = \frac{1}{2} \cdot 26 = 13$ , a nonzero constant independent of  $c$ . The Koszul involution  $c \mapsto 26 - c$  has a fixed point at  $c = 13$ , where  $\kappa(\text{Vir}_{13}) = 13/2$  and  $\text{Vir}_{13}$  is self-dual under uncurved quadratic duality.

6.4. **Genus tables.** The Faber–Pandharipande numbers and the genus expansion  $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\text{FP}}$  (uniform-weight).

TABLE 4. Faber–Pandharipande intersection numbers  $\lambda_g^{\text{FP}}$ 

| $g$ | $\lambda_g^{\text{FP}}$ | Decimal                |
|-----|-------------------------|------------------------|
| 1   | 1/24                    | 0.0416                 |
| 2   | 7/5760                  | $1.215 \times 10^{-3}$ |
| 3   | 31/967680               | $3.204 \times 10^{-5}$ |
| 4   | 127/154828800           | $8.201 \times 10^{-7}$ |
| 5   | 73/3503554560           | $2.084 \times 10^{-8}$ |

The formula is  $\lambda_g^{\text{FP}} = \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$ , where  $B_{2g}$  is the  $2g$ -th Bernoulli number. The generating function is  $\sum_{g \geq 1} \lambda_g^{\text{FP}} x^{2g} = \hat{A}(x) - 1$ , where  $\hat{A}(x) = (x/2)/\sinh(x/2)$  is the  $\hat{A}$ -genus (so  $\hat{A}(ix) = (x/2)/\sin(x/2)$ ). The numbers  $\lambda_g^{\text{FP}}$  are strictly positive, strictly decreasing, and decay like  $(2\pi)^{-2g}$ .

TABLE 5. Genus expansion  $F_g(\mathcal{A}) = \kappa \cdot \lambda_g^{\text{FP}}$  (uniform-weight) for selected families

| Algebra $\mathcal{A}$                  | $\kappa$ | $F_1$ | $F_2$    | $F_3$       |
|----------------------------------------|----------|-------|----------|-------------|
| $\mathcal{H}_1$ (Heisenberg, $k=1$ )   | 1        | 1/24  | 7/5760   | 31/967680   |
| $V_1(\mathfrak{sl}_2)$                 | 9/4      | 3/32  | 7/2560   | 31/430080   |
| $V_1(\mathfrak{sl}_3)$                 | 16/3     | 2/9   | 7/1080   | 31/181440   |
| $V_\Lambda$ (Leech, rank=24)           | 24       | 1     | 7/240    | 31/40320    |
| $\beta\gamma$ (standard, $\lambda=1$ ) | 1        | 1/24  | 7/5760   | 31/967680   |
| $\text{Vir}_1$                         | 1/2      | 1/48  | 7/11520  | 31/1935360  |
| $\text{Vir}_{13}$                      | 13/2     | 13/48 | 91/11520 | 403/1935360 |
| $\text{Vir}_{26}$                      | 13       | 13/24 | 91/5760  | 403/967680  |
| $\mathcal{W}_3$ at $c=6$               | 5        | 5/24  | 7/1152   | 31/193536   |

The entries for  $V_1(\mathfrak{sl}_3)$  use  $\kappa = (8 \cdot 4)/(2 \cdot 3) = 16/3$ . The Leech lattice (rank = 24) gives  $F_1 = 1$ , reflecting the Jacobi  $\Delta$ -function origin of the Leech theta series.

**6.5. Shadow growth rate table.** The shadow growth rate  $\rho = \sqrt{(9\alpha^2 + 16\kappa S_4)/(4\kappa^2)}$  determines the convergence of the shadow obstruction tower. For Virasoro,  $\rho^2 = (180c + 872)/[c^2(5c + 22)]$ . The critical central charge  $c^* \approx 6.125$  (the positive real root of (5.1)) separates the convergent regime ( $c > c^*$ , tower summable) from the divergent regime ( $c < c^*$ , tower grows exponentially). Class G and class L algebras have  $\rho = 0$  (the tower terminates, so there is no exponential growth).

**6.6. Virasoro shadow obstruction tower table.** Each  $S_r$  is a rational function of  $c$  with poles at  $c = 0$  and  $c = -22/5$ ; these are the two values where the quadratic Casimir  $c(5c + 22)$  vanishes. The sign pattern  $S_r > 0$  for even  $r$  and  $S_r < 0$  for odd  $r \geq 5$  (at  $c > 0$ ) reflects the oscillatory decay  $\cos(r\theta + \phi)$  from the complex branch points of the shadow generating function. All entries have been verified by two independent methods: closed-form derivation and the convolution recursion from  $H^2 = t^4 Q_L$ .

TABLE 6. Shadow growth rate  $\rho(\mathcal{A})$  for class M algebras (Virasoro at selected central charges)

| $c$                 | $\kappa$       | $Q^{\text{contact}}$ | $\rho$          | Regime     |
|---------------------|----------------|----------------------|-----------------|------------|
| 1/2 (Ising)         | 1/4            | 40/49                | $\approx 12.53$ | divergent  |
| 1                   | 1/2            | 10/27                | $\approx 6.24$  | divergent  |
| 4                   | 2              | 5/84                 | $\approx 1.54$  | divergent  |
| $c^* \approx 6.125$ | $\approx 3.06$ | $\approx 0.031$      | 1               | critical   |
| 13 (self-dual)      | 13/2           | 10/1131              | $\approx 0.467$ | convergent |
| 25                  | 25/2           | 2/735                | $\approx 0.242$ | convergent |
| 26 (bosonic string) | 13             | 5/1976               | $\approx 0.232$ | convergent |

TABLE 7. Virasoro shadow coefficients  $S_r$  for  $r = 2, \dots, 10$ 

| $r$ | $S_r(\text{Vir}_c)$                                                |
|-----|--------------------------------------------------------------------|
| 2   | $c/2$                                                              |
| 3   | 2                                                                  |
| 4   | $10/[c(5c + 22)]$                                                  |
| 5   | $-48/[c^2(5c + 22)]$                                               |
| 6   | $80(45c + 193)/[3c^3(5c + 22)^2]$                                  |
| 7   | $-2880(15c + 61)/[7c^4(5c + 22)^2]$                                |
| 8   | $80(2025c^2 + 16470c + 33314)/[c^5(5c + 22)^3]$                    |
| 9   | $-1280(2025c^2 + 15570c + 29554)/[3c^6(5c + 22)^3]$                |
| 10  | $256(91125c^3 + 1050975c^2 + 3989790c + 4969967)/[c^7(5c + 22)^4]$ |

## 7. STRUCTURAL THEOREMS

The classification is supported by four structural statements. Proposition 7.1 excludes intermediate algebraic depths; Proposition 7.2 identifies class **G** with SC-formality; and Proposition 7.4 produces class **C** by contact-channel separation. The Drinfeld–Sokolov conjecture isolates the class-raising functor under its generic-level hypotheses, and Proposition 7.8 gives the ordered classes their split direct-sum join.

**7.1. The depth gap theorem.** The Riccati equation leaves only three rank-one possibilities. The tower has support through  $S_2$  in class **G**, through  $S_3$  in class **L**, through a contact-channel  $S_4$  in class **C**, and through infinitely many nonzero coefficients in class **M**. A quadratic  $Q_L$  is either a square or a non-square; its square root is respectively polynomial or an infinite binomial series. Thus the discriminant of  $Q_L$  determines the rank-one shadow depth, while class **C** is supplied by the separate standard contact channel.

Two distinct notions of depth appear in this analysis and must be kept separate. The algebraic depth  $d_{\text{alg}}(\mathcal{A})$  counts independent shadow coefficients:  $d_{\text{alg}} = 0$  if  $S_r = 0$  for all  $r \geq 3$ ,  $d_{\text{alg}} = n$  if  $S_r = 0$  for all  $r \geq n + 3$ , and  $d_{\text{alg}} = \infty$  if infinitely many  $S_r$  are nonzero. The generating depth  $d_{\text{gen}}(\mathcal{A})$  is the degree at which higher operations are recursively determined by lower ones. For the Virasoro algebra,  $d_{\text{gen}} = 3$

(the cubic  $S_3 = 2$  and quartic  $S_4 = 10/[c(5c+22)]$  determine all  $S_r$  via the convolution recursion) but  $d_{\text{alg}} = \infty$  (every  $S_r$  is nonzero). The recursion is finitely generated; the output has infinite support.

**Proposition 7.1** (Depth gap). *Let  $\mathcal{A}$  be a chirally Koszul algebra in the standard landscape with  $\kappa(\mathcal{A}) \neq 0$ . Then*

$$d_{\text{alg}}(\mathcal{A}) \in \{0, 1, 2, \infty\}.$$

*No finite value  $d_{\text{alg}} \geq 3$  is realized. The four values correspond bijectively to the shadow classes **G**, **L**, **C**, **M**.*

*Proof.* The argument splits into a rank-one analysis on each primary line and a multi-sector extension.

*Rank one.* On a primary line  $L$  with  $\kappa \neq 0$ , the shadow generating function  $H(t) = t^2 \sqrt{Q_L(t)}$  is determined by  $(\kappa, \alpha, S_4)$ , where the shadow metric  $Q_L(t) = (2\kappa + 3\alpha t)^2 + 2\Delta t^2$  is the Gaussian decomposition and  $\Delta = 8\kappa S_4$  is the critical discriminant.

If  $\Delta = 0$  (equivalently  $S_4 = 0$ ), then  $Q_L(t) = (2\kappa + 3\alpha t)^2$  is a perfect square, and  $H(t) = t^2(2\kappa + 3\alpha t)$  is a polynomial of degree 3. The shadow coefficients  $S_r = 0$  for all  $r \geq 4$ , giving  $d_{\text{alg}} \in \{0, 1\}$  according to whether  $\alpha = 0$  (class **G**) or  $\alpha \neq 0$  (class **L**).

If  $\Delta \neq 0$  (equivalently  $S_4 \neq 0$ ), then  $Q_L$  is irreducible over the rank-one coefficient field: the interaction correction  $2\Delta t^2$  prevents  $\sqrt{Q_L}$  from being polynomial. The Taylor expansion of  $\sqrt{Q_L(t)}$  about  $t = 0$  has infinitely many nonzero coefficients (the binomial series for  $(1+u)^{1/2}$  with  $u = (6\alpha t + (9\alpha^2 + 2\Delta)t^2)/(4\kappa^2)$  has infinitely many nonzero terms when the coefficient of  $t^2$  in  $u$  is nonzero). Therefore  $S_r \neq 0$  for infinitely many  $r$ , and  $d_{\text{alg}} = \infty$  (class **M**).

The gap at  $d_{\text{alg}} = 3$  is now forced. Suppose  $S_4 \neq 0$ . The convolution recursion at degree 5 gives  $S_5 = -(6/5)(1/\kappa)\alpha S_4$ . If  $\alpha \neq 0$ : this is nonzero, so  $S_5 \neq 0$ . If  $\alpha = 0$ : then  $S_5 = 0$ , but the degree-6 recursion gives  $S_6 \propto S_4^2/\kappa \neq 0$ . In either case,  $S_4 \neq 0$  propagates to infinitely many nonzero  $S_r$ : the closed form  $H(t) = t^2 \sqrt{Q_L(t)}$  makes this manifest. Termination at any finite degree  $\geq 5$  is impossible.

Class **C** ( $d_{\text{alg}} = 2$ ) occupies the contact channel where  $S_3 = 0$ ,  $S_4 = -5/12$ , and  $S_r = 0$  for  $r \geq 5$  by a mechanism external to the rank-one Riccati analysis. See Proposition 7.4 below.

*Multi-sector extension.* For a multi-generator algebra with primary lines  $L_1, \dots, L_n$ , the cubic pump propagates: if  $\alpha \neq 0$  on any mixed sector, the adjoint  $\text{ad}_{S_3}$  generates an infinite chain on every reachable sector, forcing  $d_{\text{alg}} = \infty$ . If all mixed cubics vanish, the sectors decouple and  $d_{\text{alg}} \leq \max_i d_{\text{alg}}(L_i) \leq 2$ . No intermediate value  $d_{\text{alg}} = 3$  is accessible.  $\square$

**7.2. SC-formality characterises class G.** Every algebra in the standard landscape is chirally Koszul: the bar spectral sequence collapses at  $E_2$ , the bar cohomology concentrates in degree 1, and the transferred  $A_\infty$  operations on  $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$  satisfy  $m_k = 0$  for  $k \geq 3$ . Koszulness is common to all four shadow classes.

A stronger condition does. Swiss-cheese formality asks that the higher SC-operations on  $\mathcal{A}$  itself vanish:  $m_k^{\text{SC}} = 0$  for all  $k \geq 3$ . This is a condition on the algebra, distinct from a condition on its bar cohomology; the common notation  $m_k$  for both is a source of systematic confusion. The question is: which algebras satisfy this stronger condition?

The answer is a unique-survivor argument. Four shadow classes enter; one constraint eliminates three.

**Proposition 7.2** (SC-formality iff class **G**). *Let  $\mathcal{A}$  be a chiral algebra in the standard landscape. Then  $\mathcal{A}$  is Swiss-cheese formal if and only if  $\mathcal{A}$  belongs to class **G**.*

*Proof.* The forward direction is immediate. If  $\mathcal{A}$  is class **G** (Heisenberg, lattice, free fermion), the current algebra bracket is central:  $[a_m, a_n] = km \delta_{m+n,0}$  for Heisenberg, and the analogous statement for every other class **G** family. All nested brackets vanish, so  $m_k^{\text{SC}} = 0$  for  $k \geq 3$ .

Conversely, suppose  $\mathcal{A}$  is SC-formal. The cubic shadow  $S_3$  is the normalized projection of the trilinear form

$$C(x, y, z) = \kappa(x, [y, z]),$$

where  $\kappa$  is the invariant bilinear form and  $[-, -]$  the current algebra bracket. The chain of implications is:

$$\text{SC-formal} \implies m_3^{\text{SC}} = 0 \implies S_3 = 0 \implies C = 0.$$

The first arrow is the definition. The second holds because  $S_3$  is the degree-3 shadow projection of  $m_3^{\text{SC}}$ . The third requires the key input:  $\kappa$  is nondegenerate on the standard landscape. With  $\kappa$  nondegenerate,  $C = 0$  forces  $[y, z] = 0$  for all  $y, z$ . The current algebra is abelian; an abelian current algebra at level  $k$  is Heisenberg; Heisenberg is class **G**.

Classes **L**, **C**, and **M** all have  $S_3 \neq 0$  or carry structure (charged strata, nonlinear composites) incompatible with  $m_3^{\text{SC}} = 0$ . Each is eliminated. Only class **G** survives.  $\square$

*Remark 7.3* (The two meanings of formality). The distinction between  $\mathcal{A}_\infty$ -formality of bar cohomology and SC-formality of the algebra itself is the central subtlety of the Koszulness programme. Every chirally Koszul algebra is  $\mathcal{A}_\infty$ -formal on its bar cohomology; this is a consequence of the  $E_2$ -collapse and holds in all four shadow classes. SC-formality is a strictly stronger condition, operative on the algebra rather than on its cohomological shadow. Proposition 7.2 identifies the precise locus where the two notions coincide: class **G**.

**7.3. The betagamma contact mechanism.** The single-line Riccati dichotomy produces three outcomes:  $d_{\text{alg}} \in \{0, 1, \infty\}$ . Class **C** ( $d_{\text{alg}} = 2$ ) is absent from this list. Its existence requires a mechanism that the rank-one analysis misses: shadow data that lives on a charged stratum rather than on the primary line, and a termination mechanism that operates by dimensional obstruction rather than by algebraic identity.

The  $\beta\gamma$  system is the algebra that exhibits this mechanism. It is the unique standard family where shadow depth exceeds collision depth: the OPE pole order is  $p_{\text{max}} = 1$  (a single simple pole  $\beta(z)\gamma(w) \sim 1/(z-w)$ ), the collision depth is  $k_{\text{max}} = 0$  (no collision singularity), yet the shadow depth is  $r_{\text{max}} = 4$ . Three invariants that agree for every other standard family here separate:  $p_{\text{max}} \neq k_{\text{max}} \neq r_{\text{max}}$ . The mechanism is stratum separation: the quartic shadow lives on a charged stratum invisible to the single-line dichotomy, and the tower terminates there by rank-one rigidity.

**Proposition 7.4** (Contact class via stratum separation). *Let  $\beta\gamma_\lambda$  denote the  $\beta\gamma$  system at conformal weight  $\lambda$ , with central charge  $c(\beta\gamma_\lambda) = 2(6\lambda^2 - 6\lambda + 1)$  and  $\kappa(\beta\gamma_\lambda) = 6\lambda^2 - 6\lambda + 1$ . Then:*

- (i) *On the weight-changing primary line, the OPE is abelian ( $\alpha = 0$ ) and  $\Delta = 0$ : the single-line dichotomy gives  $r_{\text{max}}|_L = 2$  (class **G** on this line).*
- (ii) *The quartic contact invariant  $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$  is nonzero and lives in the charged sector  $\text{Def}_{\text{cyc}}^{(q)}$  with  $q \neq 0$ .*
- (iii) *The self-bracket  $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$  has  $\dim \text{Def}_{\text{cyc}}^{(2q)} = 0$  for rank-one systems: the quartic pump has zero target, and the tower terminates at  $r_{\text{max}} = 4$ .*
- (iv) *The shadow depth is  $r_{\text{max}}(\beta\gamma_\lambda) = 4$  and  $d_{\text{alg}}(\beta\gamma_\lambda) = 2$  for all  $\lambda$ .*

*Proof.* The primary-line computation gives (i), the charged-sector decomposition gives (ii), rank-one rigidity gives (iii), and (i)–(iii) imply (iv).

*Item (i): invisibility on the primary line.* The  $\beta\gamma$  OPE  $\beta(z)\gamma(w) \sim 1/(z-w)$  is a simple pole. The self-OPEs  $\beta(z)\beta(w)$  and  $\gamma(z)\gamma(w)$  are regular. There is no current algebra bracket: the OPE on the weight-changing line is abelian,  $\alpha = 0$ ,  $S_4 = 0$ ,  $\Delta = 0$ . The Riccati dichotomy applied to this line gives  $d_{\text{alg}} = 0$ . If the primary line were the whole story,  $\beta\gamma$  would be class **G**.

*Item (ii): birth of the quartic on the charged stratum.* The weight-changing primary line is incomplete. The  $\beta\gamma$  system has two generators of different conformal weight: the cyclic deformation complex decomposes by charge, and the charged sector  $\text{Def}_{\text{cyc}}^{(q)}$  with  $q \neq 0$  is invisible to the restriction  $q = 0$  that defines the primary line. The quartic contact invariant  $\mathfrak{Q}_{\beta\gamma} = \text{cyc}(m_3)$  is the trilinear Massey product on this charged stratum. In the standard conformal-weight family the quartic coefficient is  $S_4 = -5/12$  for all  $\lambda$ ; at  $\lambda = 1/2$  the symplectic boson changes the parity grading, while the graded contact witness remains nonzero.

*Item (iii): death of the quintic by rank-one rigidity.* The quartic shadow, once born, must either propagate (producing  $d_{\text{alg}} = \infty$ , class **M**) or die (giving  $d_{\text{alg}} = 2$ , class **C**). The propagation mechanism is the quartic pump: the self-bracket  $\text{Def}_{\text{cyc}}^{(q)} \otimes \text{Def}_{\text{cyc}}^{(q)} \rightarrow \text{Def}_{\text{cyc}}^{(2q)}$ . For rank-one systems (one generator in each charge sector), the target space has dimension zero: there is no field of charge  $2q$  in a rank-one system. The quintic obstruction  $S_5$  vanishes identically on the charged stratum; the pump has no room to operate. The tower terminates.

*Item (iv).* Items (i)–(iii) give  $S_r = 0$  for  $r \geq 5$  on every stratum and  $S_4 \neq 0$  on the charged stratum:  $r_{\text{max}} = 4$ ,  $d_{\text{alg}} = 2$ . The result holds for all  $\lambda$  (the rank-one rigidity is independent of conformal weight), confirming class **C**.  $\square$

*Remark 7.5* (Why contact is rare). The mechanism that produces class **C** requires three simultaneous conditions: an abelian primary line ( $\alpha = 0$ ), a nonzero quartic on a charged stratum ( $\mathfrak{Q} \neq 0$ ), and rank-one rigidity to terminate the tower ( $\dim \text{Def}^{(2q)} = 0$ ). The third condition fails for rank  $\geq 2$ : the target of the quartic pump acquires positive dimension, and the tower generically diverges. Among standard families, only the  $\beta\gamma$  system and its Koszul dual  $bc$  satisfy all three conditions.

**7.4. Shadow depth under Drinfeld–Sokolov reduction.** The four shadow classes are static invariants of individual algebras. A natural question is how these invariants transform under the functors that connect families. Quantum Drinfeld–Sokolov reduction is the principal such functor: it sends an affine Kac–Moody algebra  $V_k(\mathfrak{g})$  to a  $\mathcal{W}$ -algebra  $\mathcal{W}^k(\mathfrak{g}, f)$ , parameterised by a choice of nilpotent  $f \in \mathfrak{g}$ . At  $f = 0$  the reduction is trivial and the output is the input. At  $f \neq 0$  the output is a genuinely different algebra: new composite fields appear (the Sugawara stress tensor  $T$ , higher-spin currents  $W_j$ ), and these composites carry OPE data absent from the affine algebra.

The conjectural shadow-theoretic consequence is class escalation. An affine Kac–Moody algebra is class **L** ( $r_{\text{max}} = 3$ ): the Jacobi identity kills the quartic shadow on every primary line. For a generic nontrivial DS reduction, the stress-tensor line is required to carry a nonzero Virasoro quartic  $S_4 = 10/[c(5c+22)]$ , away from the singular central charges  $c = 0, -22/5$ . The depth gap would then force  $r_{\text{max}} = \infty$ . Singular levels and degenerate central charges require separate computation.

**Conjecture 7.6** (Shadow depth escalation under DS reduction). *Let  $\mathfrak{g}$  be a simple Lie algebra,  $f \in \mathfrak{g}$  a non-zero nilpotent, and let  $k$  be a non-critical level for which the stress-tensor line of  $\mathcal{W}^k(\mathfrak{g}, f)$  is nondegenerate and has  $c \notin \{0, -22/5\}$ . Then*

$$r_{\text{max}}(\mathcal{W}^k(\mathfrak{g}, f)) \geq r_{\text{max}}(V_k(\mathfrak{g})) = 3.$$

*The conjecture asserts strictness for every non-zero nilpotent:  $r_{\text{max}}(\mathcal{W}^k(\mathfrak{g}, f)) = \infty$  (class **M**).*

The conjecture asserts that DS reduction is a one-way door in shadow complexity: once the nilpotent  $f$  is turned on, the tower remains infinite in the conjectural generic regime. The depth gap (Proposition 7.1) reduces the conjecture to a single computation: it suffices to show  $S_4 \neq 0$  on the stress-tensor primary line of  $\mathcal{W}^k(\mathfrak{g}, f)$ . The infinite tower then follows automatically.

*Remark 7.7* (Evidence). Three independent lines of evidence support the conjecture.

(a) *Principal reductions.* For  $f = f_{\text{prin}}$  and the standard generic-level principal rows, the statement is proved: the principal  $\mathcal{W}$ -algebra  $\mathcal{W}_N = \mathcal{W}^k(\mathfrak{sl}_N, f_{\text{prin}})$  contains the Virasoro algebra as a subalgebra, and the Virasoro quartic  $S_4 \neq 0$  on the  $T$ -line is a universal computation independent of  $\mathfrak{g}$ .

(b) *Non-principal reductions in type A.* Existing computations support the conjecture for several nilpotent orbits of  $\mathfrak{sl}_N$  at non-critical levels  $k$ . The generic statement remains conjectural outside the verified OPE tables. The minimal nilpotent orbit of  $\mathfrak{sl}_3$  (the Bershadsky–Polyakov algebra  $\mathcal{W}^k(\mathfrak{sl}_3, f_{\text{min}})$ ) is the simplest non-principal example: it has a single strong generator beyond the stress tensor, and  $S_4 \neq 0$  is verified by direct OPE computation.

(c) *Structural mechanism.* Every non-trivial nilpotent  $f$  produces, via the BRST cohomology of the associated quantum DS complex, at least one composite strong generator whose self-OPE has pole order  $\geq 4$ . The pole order of the composite is bounded below by the Sugawara construction: the stress tensor  $T$  always appears as a strong generator, and  $T(z)T(w)$  has a quartic pole. This quartic pole creates  $S_4 \neq 0$  on the stress-tensor primary line, and the depth gap converts this single nonzero quartic into an infinite tower.

**7.5. The join-semilattice of shadow classes.** The four shadow classes are invariants of individual algebras. Their behaviour under algebraic operations requires a separate argument. The split operation considered here is the direct sum of cyclically admissible Lie conformal algebras with vanishing mixed OPE, whose modular envelopes split as completed tensor products.

The answer is the maximum, and the proof is that the shadow tower decomposes sector by sector when the cross-sector OPE is regular. This gives the four shadow classes the structure of a join-semilattice: a partially ordered set in which every pair of elements has a least upper bound, realised concretely by direct sum.

**Proposition 7.8** (Shadow classes form a join-semilattice). *The four shadow classes  $\{\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}\}$  form a join-semilattice under the total order  $\mathbf{G} < \mathbf{L} < \mathbf{C} < \mathbf{M}$ , where the join is realised by direct sum of chiral algebras:*

$$\text{class}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(\text{class}(\mathcal{A}_1), \text{class}(\mathcal{A}_2)).$$

*Equivalently, the shadow depth satisfies  $r_{\max}(\mathcal{A}_1 \oplus \mathcal{A}_2) = \max(r_{\max}(\mathcal{A}_1), r_{\max}(\mathcal{A}_2))$ .*

*Proof.* Assume  $\mathcal{A}_1$  and  $\mathcal{A}_2$  are cyclically admissible Lie conformal algebras with vanishing mixed OPE and split modular envelopes. The OPE of  $\mathcal{A}_1 \oplus \mathcal{A}_2$  decomposes into three sectors: pure  $\mathcal{A}_1$ , pure  $\mathcal{A}_2$ , and mixed. The mixed sector has regular OPE by hypothesis: fields from distinct factors are mutually local with no singular terms. Regular OPE contributes no shadow coefficients at any degree. The shadow tower of  $\mathcal{A}_1 \oplus \mathcal{A}_2$  is therefore the direct sum of the towers of  $\mathcal{A}_1$  and  $\mathcal{A}_2$ , and the shadow depth is the maximum of the two.

The decisive case is  $\mathbf{L} \oplus \mathbf{L}$ : does combining two affine Kac–Moody factors produce a quartic shadow? The answer is no. The quartic obstruction  $S_4$  is intrinsic to each factor’s OPE. On the primary line of  $V_k(\mathfrak{g}_1)$ , the Jacobi identity kills  $S_4$ ; on the primary line of  $V_{k'}(\mathfrak{g}_2)$ , the same. The mixed channel  $\mathfrak{g}_1 \otimes \mathfrak{g}_2$  contributes no singular OPE terms, so no quartic shadow is born in the cross-sector. Therefore  $\mathbf{L} \oplus \mathbf{L} = \mathbf{L}$ : the Jacobi identity of each factor survives the direct sum.  $\square$

**Remark 7.9** (Monotonicity and its failure). The join-semilattice property is specific to direct sums: the functor  $\oplus$  preserves the regularity of cross-sector OPE. Constructions that create new singular terms between sectors violate monotonicity and can raise the shadow class strictly. Two examples illustrate the failure.

Drinfeld–Sokolov reduction sends class  $\mathbf{L}$  affine input to class  $\mathbf{M}$  output in the verified principal rows and in the generic regime predicted by Conjecture 7.6. The BRST construction creates composite

fields with new OPE singularities; these singularities produce the quartic shadow that the Jacobi identity had killed.

Coset construction:  $\mathcal{A}/\mathcal{A}'$  can land in a higher class than either factor. The Goddard–Kent–Olive coset of  $V_k(\mathfrak{sl}_2)$  by its Cartan subalgebra produces the parafermion algebra, whose shadow depth depends on the coset data rather than on the shadow classes of the two inputs.

Shadow depth is monotone under direct sum, anti-monotone under no operation, and unconstrained under quotient or reduction. The join-semilattice records the stable part of this structure; the unstable part is the subject of Conjecture 7.6.

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*Remark 7.10* (Rank-2 holographic datum). The  $W_3$  composite field identity  $\Lambda_0|h\rangle = h^2 - 9h/5$  and the complete rank-2 holographic datum are developed in *The first rank-2 holographic datum:  $\mathcal{H}(W_3)$*  (w3\_holographic\_datum.tex).

*Remark 7.11* (Genus-two verification data). The six boundary-graph sum gives  $\partial F_2^{\text{cross}}(W_3) = (c + 204)/(16c)$ , and the Drinfeld–Sokolov BRST calculation gives Virasoro gravitational-coproduct primitivity. Exact rational evaluation at  $c \in \{1, 2, 4, 13, 26, 50, 100\}$  checks the graph table. No exceptional-Lie dimension enters the genus-2 computation. The principal Drinfeld–Sokolov statement is Theorem 9.3.

## 8. THE GENUS-2 FREE ENERGY OF $W_3$

The single-channel regime has one strong generator, one primary line, and one modular characteristic  $\kappa$ . The first multi-channel invariant occurs for  $W_3$ , whose strong generators are the stress tensor  $T$  of weight 2 and the spin-3 current  $W$  of weight 3. Thus  $F_2(W_3)$  receives both diagonal channel contributions and cross-channel contributions.

**8.1. The Frobenius algebra of the bar-complex CohFT.** The bar-complex CohFT of  $W_3$  has a two-dimensional Frobenius algebra with basis  $\{e_T, e_W\}$  and diagonal metric from the per-channel modular characteristics:

$$\eta_{TT} = \kappa_T = \frac{c}{2}, \quad \eta_{WW} = \kappa_W = \frac{c}{3}, \quad \eta_{TW} = 0.$$

The total modular characteristic is  $\kappa(W_3) = \kappa_T + \kappa_W = 5c/6 = c \cdot (H_3 - 1)$ , where  $H_3 = 1 + 1/2 + 1/3 = 11/6$  is the third harmonic number. The inverse metric gives the propagators:

$$\eta^{TT} = \frac{2}{c}, \quad \eta^{WW} = \frac{3}{c}.$$

Both propagators have weight 1: the bar-complex propagator is  $d \log E(z, w)$ , which has weight 1 regardless of the conformal weight of the field being sewed.

The OPE determines the three-point structure constants. The  $\mathbb{Z}_2$  symmetry  $W \mapsto -W$  kills all three-point functions with an odd number of  $W$ -insertions:

$$C_{TTT} = C_{TWW} = C_{WWT} = c, \quad C_{TTW} = C_{TWT} = C_{WTT} = C_{WWW} = 0.$$

The OPE identities give the constants:  $T_{(1)}T = 2T$  gives  $C_{TT}^T = 2$ , whence  $C_{TTT} = \eta_{TT} \cdot 2 = c$ ;  $T_{(1)}W = 3W$  gives  $C_{TW}^W = 3$ , whence  $C_{TWW} = \eta_{WW} \cdot 3 = c$ ;  $W_{(3)}W = 2T$  gives  $C_{WW}^T = 2$ , whence  $C_{WWT} = \eta_{TT} \cdot 2 = c$ .

The genus-0 four-point vertex is universal: for any channel pair  $(i, j) \in \{T, W\}^2$ ,

$$V_{0,4}(i, i \mid j, j) = \sum_m \eta^{mm} C_{iim} C_{jjm} = \eta^{TT} \cdot c \cdot c = 2c. \quad (8.1)$$

Only the  $T$ -channel contributes to the sum:  $C_{iiT} = c$  for  $i \in \{T, W\}$ , while  $C_{iiW} = 0$  for both  $i = T$  (since  $C_{TTW} = 0$ ) and  $i = W$  (since  $C_{WWW} = 0$ ). The  $W$ -channel is invisible at genus-0 valence 4.

**8.2. The seven stable graphs at genus 2.** The moduli space  $\overline{\mathcal{M}}_{2,0}$  has a stratification by seven stable graphs  $\Gamma_0, \dots, \Gamma_6$ :

|            | Name         | Vertices                                    | Edges                   | Aut | $h^1$ | Type            |
|------------|--------------|---------------------------------------------|-------------------------|-----|-------|-----------------|
| $\Gamma_0$ | smooth       | $(g=2, \text{val}=0)$                       | 0                       | 1   | 0     | compact stratum |
| $\Gamma_1$ | figure-eight | $(g=1, \text{val}=2)$                       | 1 self-loop             | 2   | 1     | one-loop        |
| $\Gamma_2$ | banana       | $(g=0, \text{val}=4)$                       | 2 self-loops            | 8   | 2     | two-loop sunset |
| $\Gamma_3$ | dumbbell     | $(g=1, \text{val}=1) + (g=1, \text{val}=1)$ | 1 bridge                | 2   | 0     | separating      |
| $\Gamma_4$ | theta        | $(g=0, \text{val}=3) + (g=0, \text{val}=3)$ | 3 bridges               | 12  | 2     | non-separating  |
| $\Gamma_5$ | tadpole      | $(g=0, \text{val}=3) + (g=1, \text{val}=1)$ | 1 self-loop + 1 bridge  | 2   | 1     | mixed           |
| $\Gamma_6$ | chain        | $(g=0, \text{val}=3) + (g=0, \text{val}=3)$ | 2 self-loops + 1 bridge | 8   | 2     | non-separating  |

Every vertex satisfies the stability condition  $2g_v + \text{val}_v \geq 3$ , and the genus formula  $h^1(\Gamma) + \sum_v g_v = 2$  holds in each case. The smooth graph  $\Gamma_0$  has no edges and contributes the integrated class on  $\mathcal{M}_{2,0}$  itself; the remaining six graphs are the boundary strata.

**8.3. Multi-channel Feynman rules.** For each graph  $\Gamma$  with  $|E(\Gamma)|$  edges, a *channel assignment* is a function  $\sigma: E(\Gamma) \rightarrow \{T, W\}$ . The Feynman rules of the bar-complex CohFT assign to each pair  $(\Gamma, \sigma)$  an amplitude

$$\mathcal{A}(\Gamma, \sigma) = \frac{1}{|\text{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e), \sigma(e)} \prod_{v \in V(\Gamma)} V_v(g_v; \sigma|_{\text{half-edges}(v)}),$$

where the vertex factors are:

- *Genus-0, trivalent*:  $V_v = C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$ .
- *Genus-0, four-valent* (the banana vertex):  $V_v = \sum_m \eta^{mm} C_{\sigma(e_1)\sigma(e_2)m} C_{m\sigma(e_3)\sigma(e_4)}$ .
- *Genus-1, valence  $n$* :  $V_v = \kappa_{\sigma(e)}/24$  (per-channel genus-1 universality).

Each half-edge at a self-loop contributes a pair of half-edges to the same vertex, both carrying channel  $\sigma(e)$ .

**8.4. Per-graph amplitudes.** Each boundary graph amplitude decomposes into *diagonal* contributions, where all edges lie in one channel, and *mixed* contributions, where both channels occur.

$\Gamma_1$ : *figure-eight*. One edge, two channel assignments. The vertex is genus-1 with valence 2, so the vertex factor for channel  $i$  is  $\kappa_i/24$ :

$$\begin{aligned} \mathcal{A}(\Gamma_1, T) &= \frac{1}{2} \eta^{TT} \cdot \frac{\kappa_T}{24} = \frac{1}{2} \cdot \frac{2}{c} \cdot \frac{c}{48} = \frac{1}{48}, \\ \mathcal{A}(\Gamma_1, W) &= \frac{1}{2} \eta^{WW} \cdot \frac{\kappa_W}{24} = \frac{1}{2} \cdot \frac{3}{c} \cdot \frac{c}{72} = \frac{1}{48}. \end{aligned}$$

Both channels contribute equally: the propagator  $1/\kappa_i$  cancels the genus-1 vertex factor  $\kappa_i/24$ , leaving the universal value  $1/48$  independent of  $c$ . The total is  $1/24 = \lambda_1^{\text{FP}}$ . No mixed channel assignments exist (single edge).

$\Gamma_2$ : *banana*. Two edges, four channel assignments,  $|\text{Aut}| = 8$ . The vertex is genus-0 with valence 4, and the four-point vertex factor is  $V_{0,4}(i, i \mid j, j) = 2c$  universally (8.1).

$$\begin{aligned}\mathcal{A}(\Gamma_2, TT) &= \frac{1}{8} (\eta^{TT})^2 \cdot 2c = \frac{1}{8} \cdot \frac{4}{c^2} \cdot 2c = \frac{1}{c}, \\ \mathcal{A}(\Gamma_2, WW) &= \frac{1}{8} (\eta^{WW})^2 \cdot 2c = \frac{1}{8} \cdot \frac{9}{c^2} \cdot 2c = \frac{9}{4c}, \\ \mathcal{A}(\Gamma_2, TW+WT) &= \frac{1}{8} \cdot 2 \cdot \eta^{TT} \eta^{WW} \cdot 2c = \frac{1}{8} \cdot 2 \cdot \frac{6}{c^2} \cdot 2c = \frac{3}{c}.\end{aligned}\quad (8.2)$$

The cross-channel contribution is  $\delta_{\text{banana}} = 3/c$ .

$\Gamma_3$ : *dumbbell*. One bridge edge, two channel assignments,  $|\text{Aut}| = 2$ . Each vertex has genus 1 and valence 1, with vertex factor  $\kappa_i/24$ :

$$\begin{aligned}\mathcal{A}(\Gamma_3, T) &= \frac{1}{2} \eta^{TT} \cdot \left(\frac{\kappa_T}{24}\right)^2 = \frac{1}{2} \cdot \frac{2}{c} \cdot \frac{c^2}{2304} = \frac{c}{2304}, \\ \mathcal{A}(\Gamma_3, W) &= \frac{1}{2} \eta^{WW} \cdot \left(\frac{\kappa_W}{24}\right)^2 = \frac{1}{2} \cdot \frac{3}{c} \cdot \frac{c^2}{5184} = \frac{c}{3456}.\end{aligned}$$

The total is  $c/2304 + c/3456 = (3c + 2c)/6912 = 5c/6912 = \kappa(W_3)/1152$ . No mixed channel assignments exist (single edge).

$\Gamma_4$ : *theta*. Three bridge edges, eight channel assignments,  $|\text{Aut}| = 12$ . Each vertex has genus 0 and valence 3, with vertex factor  $C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$ .

- *All-T*: both vertices get  $C_{TTT} = c$ . Amplitude  $= \frac{1}{12} (\eta^{TT})^3 \cdot c^2 = \frac{1}{12} \cdot \frac{8}{c^3} \cdot c^2 = \frac{2}{3c}$ .
- *All-W*: both vertices get  $C_{WWW} = 0$ . Amplitude  $= 0$ .
- *Type (T, T, W)* (three such assignments): each vertex receives one  $W$  half-edge, giving  $C_{TTW} = 0$ . All three vanish.
- *Type (T, W, W)* (three such assignments): each vertex receives half-edges  $(T, W, W)$ , giving  $C_{TWW} = c$ . The propagator product is  $\eta^{TT} (\eta^{WW})^2 = (2/c)(9/c^2) = 18/c^3$ . Each assignment contributes  $18c^2/c^3 = 18/c$ ; with  $|\text{Aut}| = 12$ , the total mixed amplitude is

$$\delta_{\text{theta}} = \frac{3 \cdot 18}{12c} = \frac{9}{2c}. \quad (8.3)$$

$\Gamma_5$ : *tadpole*. One self-loop at vertex  $v_0$  (genus 0, valence 3) and one bridge to vertex  $v_1$  (genus 1, valence 1). Two edges, four channel assignments,  $|\text{Aut}| = 2$ .

- $(T, T)$ : self-loop  $T$ , bridge  $T$ . Vertex  $v_0$  gets half-edges  $(T, T, T) \rightarrow C_{TTT} = c$ . Vertex  $v_1$  gets half-edge  $T \rightarrow \kappa_T/24 = c/48$ . Propagators:  $(\eta^{TT})^2 = 4/c^2$ . With  $|\text{Aut}|: \frac{1}{2} \cdot \frac{4}{c^2} \cdot c \cdot \frac{c}{48} = \frac{1}{24}$ .
- $(W, W)$ : self-loop  $W$ , bridge  $W$ . Vertex  $v_0$  gets  $(W, W, W) \rightarrow C_{WWW} = 0$ . Vanishes.
- $(T, W)$ : self-loop  $T$ , bridge  $W$ . Vertex  $v_0$  gets  $(T, T, W) \rightarrow C_{TTW} = 0$ . Vanishes.
- $(W, T)$ : self-loop  $W$ , bridge  $T$ . Vertex  $v_0$  gets  $(W, W, T) \rightarrow C_{WWT} = c$ . Vertex  $v_1$  gets  $T \rightarrow \kappa_T/24 = c/48$ . Propagators:  $\eta^{WW} \cdot \eta^{TT} = (3/c)(2/c) = 6/c^2$ . With  $|\text{Aut}|:$

$$\delta_{\text{tadpole}} = \frac{1}{2} \cdot \frac{6}{c^2} \cdot c \cdot \frac{c}{48} = \frac{6c^2}{96c^2} = \frac{1}{16}. \quad (8.4)$$

The tadpole mixed amplitude  $1/16$  is *independent of  $c$* : the  $c^2$  in the numerator (from  $C_{WWT} \cdot \kappa_T/24$ ) cancels the  $c^2$  in the denominator (from  $\eta^{WW} \cdot \eta^{TT}$ ).

$\Gamma_6$ : *chain (barbell)*. Two genus-0 trivalent vertices, each carrying one self-loop, connected by a bridge. Three edges (self-loop  $s_1$ , self-loop  $s_2$ , bridge  $b$ ), eight channel assignments,  $|\text{Aut}| = 8$ . Each vertex  $v_j$  receives half-edges  $(\sigma(s_j), \sigma(s_j), \sigma(b))$  with vertex factor  $C_{\sigma(s_j) \sigma(s_j) \sigma(b)}$ .

Only  $\sigma(b) = T$  gives nonzero vertex factors (since  $C_{TTW} = C_{WWW} = 0$ ). Among the four remaining assignments  $(\sigma(s_1), \sigma(s_2)) \in \{T, W\}^2$  with  $\sigma(b) = T$ :

- $(T, T, T)$ : diagonal. Vertex factors  $C_{TTT}^2 = c^2$ , propagators  $(\eta^{TT})^3 = 8/c^3$ . With  $|\text{Aut}|$ :  $(1/8)(8/c^3)(c^2) = 1/c$ .
- $(T, W, T)$ : mixed.  $C_{TTT} \cdot C_{WWT} = c^2$ ,  $\eta^{TT} \eta^{WW} \eta^{TT} = 12/c^3$ . With  $|\text{Aut}|$ :  $(1/8)(12/c^3)(c^2) = 3/(2c)$ .
- $(W, T, T)$ : mixed. By the  $s_1 \leftrightarrow s_2$  symmetry:  $3/(2c)$ .
- $(W, W, T)$ : mixed.  $C_{WWT}^2 = c^2$ ,  $(\eta^{WW})^2 \eta^{TT} = 18/c^3$ . With  $|\text{Aut}|$ :  $(1/8)(18/c^3)(c^2) = 9/(4c)$ .

The total mixed amplitude is

$$\delta_{\text{barbell}} = \frac{3}{2c} + \frac{3}{2c} + \frac{9}{4c} = \frac{6+6+9}{4c} = \frac{21}{4c}. \quad (8.5)$$

### 8.5. The cross-channel correction.

**Proposition 8.1** (Rank-2 genus-two cross-channel term). *For the  $W_3$  bar-complex CohFT with the Feynman rules above, the degree-0 mixed-boundary graph sum is*

$$\delta F_2^{\text{cross}}(W_3) = \underbrace{\frac{3}{c}}_{\text{banana}} + \underbrace{\frac{9}{2c}}_{\text{theta}} + \underbrace{\frac{1}{16}}_{\text{tadpole}} + \underbrace{\frac{21}{4c}}_{\text{barbell}} = \frac{51}{4c} + \frac{1}{16} = \frac{c + 204}{16c}.$$

*Proof.* The mixed columns of the six boundary-graph amplitudes contribute only on the banana, theta, tadpole, and barbell graphs. Summing Equations (8.2), (8.3), (8.4), and (8.5) gives the displayed rational function. The figure-eight and dumbbell graphs have a single edge, hence no mixed channel assignment.  $\square$

The figure-eight and dumbbell graphs contribute zero mixed amplitude (single edges admit only diagonal assignments). The correction decomposes into two structurally distinct pieces:

- A *c*-dependent piece  $51/(4c)$ , arising from genus-0 vertex factors on the banana ( $3/c$ ), theta ( $9/(2c)$ ), and barbell ( $21/(4c)$ ) graphs. This is *R*-matrix independent: genus-0 vertices receive no corrections from the CohFT *R*-matrix.
- A *c*-independent piece  $1/16$ , arising from the tadpole graph, which mixes a genus-0 vertex factor with a genus-1 vertex factor  $\kappa_i/24$ .

**Remark 8.2** (Comparison with the scalar approximation). The per-channel (diagonal) graph sum, computed by applying Theorem D to each channel independently, gives

$$F_2^{\text{scal}}(W_3) = \kappa \cdot \lambda_2^{\text{FP}} = \frac{5c}{6} \cdot \frac{7}{5760} = \frac{7c}{6912}.$$

The ratio of the cross-channel correction to the scalar approximation is

$$\frac{\delta F_2^{\text{cross}}}{F_2^{\text{scal}}} = \frac{432(c + 204)}{7c^2},$$

which diverges as  $c \rightarrow 0$  and decays as  $c^{-1}$  at large  $c$ . At  $c = 4$ : the ratio is approximately 802, and the cross-channel correction *dominates* the scalar term. At  $c = 50$ : the ratio is approximately 6.3. At large  $c$ , the tadpole's *c*-independent contribution  $1/16$  dominates, and the correction becomes a subleading  $O(1/c)$  effect relative to the  $O(c)$  scalar term.

*Remark 8.3* (Cross-channel correction). The graph computation uses the genus-1 vertex factors  $\kappa_i/24$  before the CohFT  $R$ -matrix action. The multi-generator universality problem has a negative answer: the CohFT  $R$ -matrix contributes no terms cancelling  $\partial F_2^{\text{cross}}$ . Hence  $\partial F_2^{\text{cross}} = (c + 204)/(16c)$  is a genuine additional term, and the decomposition is  $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \partial F_g^{\text{cross}}$ .

**8.6. The complete amplitude table.** The full per-graph data are:

| Graph                   | Diagonal ( $T$ ) | Diagonal ( $W$ ) | Mixed     | Total     |
|-------------------------|------------------|------------------|-----------|-----------|
| $\Gamma_1$ figure-eight | $1/48$           | $1/48$           | $0$       | $1/24$    |
| $\Gamma_2$ banana       | $1/c$            | $9/(4c)$         | $3/c$     | $25/(4c)$ |
| $\Gamma_3$ dumbbell     | $c/2304$         | $c/3456$         | $0$       | $5c/6912$ |
| $\Gamma_4$ theta        | $2/(3c)$         | $0$              | $9/(2c)$  | $31/(6c)$ |
| $\Gamma_5$ tadpole      | $1/24$           | $0$              | $1/16$    | $5/48$    |
| $\Gamma_6$ barbell      | $1/c$            | $0$              | $21/(4c)$ | $25/(4c)$ |

The table records the six boundary-graph amplitudes; the smooth graph  $\Gamma_0$  has no edges and contributes an integrated class on  $\mathcal{M}_{2,0}$  itself. The mixed column sums to  $\partial F_2^{\text{cross}} = (c + 204)/(16c)$ .

The rational identities in the table specialise correctly under exact arithmetic at  $c \in \{1, 2, 4, 13, 26, 50, 100\}$ .

**8.7. Planted-forest corrections.** The rank-one planted-forest correction on a single primary line is  $\partial_{\text{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$ . On the Virasoro  $T$ -line,  $S_3^T = 2$  and  $\kappa_T = c/2$ , hence

$$\partial_{\text{pf}}^{T, \text{rank} 1} = \frac{2(20 - c/2)}{48} = \frac{40 - c}{48}.$$

The pure  $W$ -line rank-one projection has  $\partial_{\text{pf}}^{W, \text{pure}} = 0$  because the  $\mathbb{Z}_2$  parity  $W \mapsto -W$  kills odd-degree shadows on the pure  $W$ -line. The full rank-2  $W_3$  correction also contains the mixed cubic  $S_3(T, W, W) = 1$ , and the canonical rank-2 value is

$$\partial_{\text{pf}}^{(2,0)}(W_3) = \frac{932 - 7c^2}{48c^3}.$$

**8.8. Koszul duality constraints.** Under the Koszul involution,  $W_3$  at central charge  $c$  is dual to  $W_3$  at central charge  $c' = 100 - c$  (the Feigin–Frenkel involution for  $\mathfrak{sl}_3$ ). The modular characteristics satisfy

$$\kappa(c) + \kappa(100 - c) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{250}{3},$$

and the scalar free energies satisfy  $F_2^{\text{scal}}(c) + F_2^{\text{scal}}(100 - c) = (250/3) \cdot \lambda_2^{\text{FP}}$ . The cross-channel correction at the dual pair is

$$\partial F_2(c) + \partial F_2(100 - c) = \frac{c + 204}{16c} + \frac{304 - c}{16(100 - c)} = -\frac{c^2 - 100c - 10200}{8c(100 - c)}.$$

This sum varies with  $c$ : the cross-channel correction violates Koszul additivity. Since the  $R$ -matrix contributes no cancellation term,  $F_2$  contains a Koszul-asymmetric contribution beyond the scalar single-channel term.

## 9. THE GRAVITATIONAL COPRODUCT IS PRIMITIVE

In gauge theory, gluons split: one gluon can exit a scattering region as two, and the dg-shifted Yangian coproduct encodes this splitting. In gravity, the graviton is composite. The stress tensor is the Sugawara quadratic

$$T = \frac{1}{k+2} \left( \frac{1}{2} :J^0 J^0: + :J^+ J^-: \right),$$

built from affine currents by normal ordering. This compositeness has a precise algebraic consequence: the transferred coproduct is trivial.

**9.1. The DS-HPL transfer theorem.** The Drinfeld–Sokolov reduction  $\hat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$  is a BRST reduction with a strong deformation retract  $(i, p, h)$  from the BRST complex to the Virasoro cohomology. The homotopy  $h$  has ghost-number degree  $-1$ : it maps  $V \mapsto b$  (ghost number  $0 \rightarrow -1$ ) and  $c \mapsto U/2$  (ghost number  $1 \rightarrow 0$ ).

**Construction 9.1** (DS-HPL transfer data). Let  $Y^{\text{dg}}(\hat{\mathfrak{sl}}_2)$  denote the dg-shifted Yangian at level  $k$ , with  $A_\infty$  products  $\{m_n^{\text{aff}}\}$ , coproduct  $\Delta_z^{\text{aff}}$ , and  $r$ -matrix  $r^{\text{aff}, \text{coll}}(z) = k\Omega_{\text{tr}}/z$  in the trace-form collision normalization. The HPL transfer through the BRST SDR produces:

- (i) Transferred  $A_\infty$  products  $\{m_n^{\text{Vir}}\}_{n \geq 2}$  on the Virasoro sector, with  $m_n^{\text{Vir}} \neq 0$  for all  $n \geq 2$  (the entire infinite  $A_\infty$  tower of class **M**).
- (ii) A transferred coproduct  $\Delta_z^{\text{Vir}}: H_{\text{Vir}} \rightarrow H_{\text{Vir}} \otimes H_{\text{Vir}}$ .
- (iii) A transferred  $r$ -matrix  $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$ , with cubic leading pole reflecting the Sugawara nonlinearity.

The trace-form affine collision  $r$ -matrix  $r^{\text{aff}, \text{coll}}(z) = k\Omega_{\text{tr}}/z$  has a simple pole: the OPE has at most a double pole, and the bar kernel  $d\log(z-w)$  absorbs one power. The KZ-normalized connection uses  $r^{\text{KZ}}(z) = \Omega_{\text{KZ}}/((k+2)z)$  for  $\mathfrak{sl}_2$ , with a different normalization of the Casimir. The Virasoro OPE has a quartic pole ( $z^{-4}$  in  $T(z)T(w)$ ); absorption produces the cubic pole  $z^{-3}$  of  $r^{\text{Vir}}(z)$ . The general principle is:  $r$ -matrix pole order equals OPE pole order minus one.

**Theorem 9.2** (Virasoro coproduct primitivity). *For the verified Virasoro DS reduction, the HPL-transferred coproduct is strictly primitive at all degrees:*

$$\Delta_{z,n}^{\text{Vir}} = 0 \quad \text{for all } n \geq 2.$$

That is,  $\Delta_z^{\text{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T$ . All nontrivial structure of the Virasoro dg-shifted Yangian resides in the  $r(z)$ -twisted target product, never in the coproduct.

*Proof.* The tensor-factor projection formulation of the Virasoro DS complex gives the result.

*Linear term.* The linear transferred coproduct  $(p \otimes p)(\Delta_z(T))$  is already primitive. The affine coproduct  $\Delta_z(T)$  expands as  $\tau_z(T) \otimes 1 + 1 \otimes T + \frac{1}{k+2} \Omega(w+z, w)$ , where  $\Omega(w+z, w) = J^0(w+z) \otimes J^0(w) + J^+(w+z) \otimes J^-(w) + J^-(w+z) \otimes J^+(w)$ . The projection  $p$  annihilates  $J^0$  (outside the Virasoro sector) and  $J^+$  decomposes as  $V + 1$  with  $p(V) = 0$ . Every summand of  $\Omega$  contains at least one factor from  $\{J^0, J^-, J^+\}$  at conformal weight 1, where the Virasoro sector has no states ( $L_{-1}|0\rangle = 0$ ). Each tensor factor of  $(p \otimes p)(\Omega)$  has  $p$  killing at least one component, so  $(p \otimes p)(\Omega) = 0$ .

*Higher HPL trees.* The degree-2 source correction has the form

$$(p \otimes p)\Delta_z h \mu(iT, iT),$$

and the degree-2 target correction has the form

$$(p \otimes p)H_{12} \mu_{r(z)}(\Delta_z iT, \Delta_z iT).$$

The source tree lands in the BRST ghost sector after the homotopy  $h$ ; the target tree lands in the same ghost sector after the  $r(z)$ -twisted target product. The projection  $p \otimes p$  kills both. In higher degrees, each transferred source or target tree contains the same source/target ghost split: either a homotopy edge places the output in nonzero ghost degree, or the  $r(z)$ -twisted target product does. Thus every higher coproduct component is annihilated by  $p \otimes p$ .

The ghost degrees are:

| Operation                      | Ghost degree |
|--------------------------------|--------------|
| $h$ (homotopy)                 | $-1$         |
| $\delta$ (BRST perturbation)   | $+1$         |
| $\mu$ (vertex algebra product) | $0$          |
| $\Delta_z$ (coproduct)         | $0$          |

*Projection to cohomology.* The  $W$ -algebra cohomology sits in ghost number  $0$ . The projection  $p$  annihilates all ghost-number ( $\neq 0$ ) elements. The projection ( $p \otimes p$ ) therefore kills every source and target correction in degree  $n \geq 2$ .

The perturbation produces no corrections. The BRST perturbation  $\delta$  has ghost degree  $+1$ , so the product  $h\delta$  preserves ghost number. But  $\delta(T) = 0$  (the stress tensor is BRST-closed), so  $(h\delta)(i(T)) = 0$ , and by induction  $(h\delta)^n(i(T)) = 0$  for all  $n \geq 1$ . The corrected inclusion  $i_\infty(T) = i(T)$  is unperturbed, and the same primitive coproduct results at every perturbative order.  $\square$

**Theorem 9.3** (Principal DS coproduct primitivity). *For every principal Drinfeld–Sokolov reduction  $V_k(\mathfrak{g}) \rightarrow \mathcal{W}_k(\mathfrak{g})$ ,*

$$\Delta_{z,n}^W = 0 \quad (n \geq 2), \quad \Delta_z^W(x) = \tau_z(x) \otimes 1 + 1 \otimes x.$$

*Proof.* The principal DS BRST complex has the same source/target ghost split as the Virasoro case. The linear affine coproduct projects to the primitive term on cohomology. Each higher transferred source tree contains a homotopy edge landing in nonzero ghost degree, and each higher transferred target tree contains the corresponding  $r(z)$ -twisted target product in nonzero ghost degree. The principal projection kills these components, leaving only  $\tau_z(x) \otimes 1 + 1 \otimes x$ .  $\square$

**9.2. Physical meaning.** The coproduct  $\Delta_z$  of a dg-shifted Yangian encodes the *splitting* of excitations. Its strictness or nontriviality is the algebraic fingerprint of the distinction between fundamental and composite fields.

| Theory             | Boundary algebra                                | $\Delta_z$                | Shadow depth               | Reason                   |
|--------------------|-------------------------------------------------|---------------------------|----------------------------|--------------------------|
| Chern–Simons       | $\hat{\mathfrak{g}}_k$ (affine KM)              | reduced coproduct nonzero | 3 (class <b>L</b> )        | fundamental gluons split |
| 3d gravity         | $\text{Vir}_c = \mathcal{W}_k(\mathfrak{sl}_2)$ | $\Delta_{z,n \geq 2} = 0$ | $\infty$ (class <b>M</b> ) | ghost obstruction (DS)   |
| Twisted holography | $H_k$ (Heisenberg)                              | $\Delta_{z,n \geq 2} = 0$ | 2 (class <b>G</b> )        | free field, $r = k/z$    |
| M2 brane           | DDCA                                            | reduced coproduct nonzero | non-DS                     | no DS, fundamental modes |
| M5 brane           | $\mathcal{W}_{1+\infty}(K)$                     | $\Delta_{z,n \geq 2} = 0$ | $\infty$ (class <b>M</b> ) | ghost obstruction (DS)   |

Theorem 9.3 states the principal Drinfeld–Sokolov result: the BRST ghost grading kills all higher coproduct components. Theorem 9.2 is its Virasoro instance for  $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$ . Theories with fundamental excitations (Chern–Simons, deformed double current algebras) lack a ghost sector, and their coproducts are nontrivial.

The two exceptions to the DS pattern are illuminating. The Heisenberg algebra has a primitive coproduct for a different reason: it is a free field, its  $r$ -matrix  $r^H(z) = k/z$  is scalar, and the bar complex

is formal at degree 2 (shadow depth  $r_{\max} = 2$ , class **G**). No DS reduction and no ghost obstruction; primitivity follows from the absence of nonlinear OPE structure. The M2 brane boundary algebra (a deformed double current algebra) has a nontrivial coproduct because it lies outside the DS-reduction family; its excitations are fundamental modes of the M2 worldvolume theory.

**9.3. The  $r$ -matrix carries all gravitational structure.** With the coproduct primitive, the full gravitational content of the Virasoro dg-shifted Yangian is encoded in the  $r$ -matrix

$$r^{\text{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}.$$

The two terms encode distinct physical data:

- The *leading coefficient*  $c/2 = \kappa(\text{Vir}_c)$  is the modular characteristic. It determines the uniform-weight genus tower  $F_g = (c/2) \int_{\overline{M}_g} \lambda_g$  and the two-sided BTZ entropy

$$S_{\text{BTZ}} = 2\pi \sqrt{\frac{c\Delta_+}{6}} + 2\pi \sqrt{\frac{c\Delta_-}{6}}, \quad \Delta_{\pm} = L_{\text{o}}^{\pm} - c/24.$$

- The *subleading term*  $2T/z$  encodes the state dependence of gravitational scattering. The braiding of Wilson lines (gravitational line operators) in the holomorphic-topological bulk is controlled by this term.
- The *pole structure* (cubic leading pole, simple subleading pole, no even poles) is the signature of a bosonic algebra extracted through the  $d\log$  kernel: the OPE poles  $z^{-4}, z^{-2}, z^{-1}$  each lose one order under the  $d\log$  extraction, producing  $r$ -matrix terms  $z^{-3}, z^{-1}, z^0$ . The  $z^0$  term (from the  $\partial T$  pole at  $z^{-1}$ ) is regular and contributes no term to the  $r$ -matrix, leaving only  $z^{-3}$  and  $z^{-1}$ .

For Chern–Simons gauge theory, the trace-form affine collision  $r$ -matrix

$$r^{\text{aff}, \text{coll}}(z) = \frac{k\Omega_{\text{tr}}}{z}$$

has a simple pole and the reduced coproduct is nonzero. For gravity, the cubic-pole Virasoro  $r$ -matrix controls braiding while the coproduct remains primitive. Drinfeld–Sokolov reduction sends the affine collision datum to  $(c/2)/z^3 + 2T/z$  by homological perturbation through the Sugawara nonlinearity; the ghost-number obstruction annihilates the nonprimitive coproduct terms.

| Datum                        | Chern–Simons                                     | Gravity                           |
|------------------------------|--------------------------------------------------|-----------------------------------|
| boundary algebra             | $V_k(\mathfrak{g})$ (affine KM)                  | $\text{Vir}_c$ (Virasoro)         |
| highest OPE pole             | $z^{-2}$ (quadratic)                             | $z^{-4}$ (quartic)                |
| $\mathcal{A}_{\infty}$ tower | strict ( $m_{k \geq 3} = 0$ )                    | infinite ( $m_k \neq 0$ all $k$ ) |
| shadow depth                 | 3 (class <b>L</b> )                              | $\infty$ (class <b>M</b> )        |
| $r^{\text{coll}}(z)$         | $k\Omega_{\text{tr}}/z$ (simple pole)            | $(c/2)/z^3 + 2T/z$ (cubic pole)   |
| $\Delta_z$                   | nontrivial (gluons split)                        | primitive (gravitons braid)       |
| $\kappa$                     | $\dim \mathfrak{g} (k + h^{\vee}) / (2h^{\vee})$ | $c/2$                             |
| $\kappa + \kappa^!$          | 0                                                | 13                                |

The entire table descends from a single datum: the number of  $\lambda$ -powers in the  $\lambda$ -bracket. Affine Kac–Moody has  $\lambda^1$  at most (quadratic pole), yielding strict associativity, finite shadow depth, a simple-pole  $r$ -matrix, and a nontrivial coproduct. The Virasoro algebra has  $\lambda^3$  (quartic pole), yielding the full infinite  $\mathcal{A}_{\infty}$  tower, infinite shadow depth, a cubic-pole  $r$ -matrix, and a primitive coproduct. The Drinfeld–Sokolov reduction is the functor that connects the two columns: it increases the  $\lambda$ -bracket

order by two, trades the nontrivial coproduct for a ghost obstruction, and replaces the complementarity  $\kappa + \kappa^\dagger = 0$  with  $\kappa + \kappa^\dagger = 13$ .

Thus the Virasoro dg-shifted Yangian has a primitive coproduct and a nontrivial cubic-pole  $r$ -matrix: the composite Sugawara origin of  $T$  places gravitational interaction in braiding, with no higher coproduct component.

## 10. OPEN PROBLEMS AND OUTLOOK

We collect several directions suggested by the results of this paper.

**10.1. Multi-generator universality at genus  $\geq 2$ .** The genus-2 computation for  $\mathcal{W}_3$  (Section 8) produces a cross-channel correction  $\delta F_2 = (c + 204)/(16c)$  to the scalar approximation  $F_2^{\text{scal}} = \kappa \cdot \lambda_2^{\text{FP}}$ . This correction is  $R$ -matrix independent and is not cancelled by CohFT  $R$ -matrix terms: the multi-generator universality problem is resolved negatively at genus 2. The correct decomposition is  $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$ . At genus 1, the identity  $F_1 = \kappa \cdot \lambda_1^{\text{FP}}$  holds unconditionally for all families. The structure of  $\delta F_g^{\text{cross}}$  at genus  $g \geq 3$  remains to be computed.

**10.2. The operadic complexity conjecture.** The four-class partition **G/L/C/M** classifies shadow depth. The operadic complexity theorem identifies  $r_{\max}(\mathcal{A})$  with the  $A_\infty$ -depth (the minimal  $n$  such that  $m_k = 0$  for  $k > n$  in the transferred  $A_\infty$  structure) and with the  $L_\infty$ -formality level. The shadow-formality identification is proved at all degrees by induction on  $r$  (thm:shadow-formality-identification).

**10.3. Pixton's ideal and the shadow obstruction tower.** For class **M** algebras (infinite shadow obstruction tower), the Maurer–Cartan equation at each genus  $g$  produces a tautological relation on  $\overline{\mathcal{M}}_g$ . These relations lie in the Pixton ideal (the ideal of relations among tautological classes), and it is natural to ask whether the shadow obstruction tower of a single class **M** algebra generates the full ideal. Computational evidence at genus 2 and 3 supports this; a proof would require establishing ideal membership in the strata algebra.

**10.4. Arithmetic content of shadow denominators.** The Virasoro shadow coefficients  $S_r$  are rational functions of  $c$  with poles at  $c = 0$  and  $c = -22/5$ . The denominators  $c^{r-3}(5c + 22)^{\lfloor (r-2)/2 \rfloor}$  grow in a pattern controlled by the Gaussian decomposition. For degrees  $r \geq 12$ , the numerators may be divisible by Ramanujan-type primes whose appearance is governed by cusp forms. The total depth decomposition  $d(\mathcal{A}) = 1 + d_{\text{arith}} + d_{\text{alg}}$  separates the algebraic engine (depths  $\leq 4$ ) from the arithmetic frontier (depths  $\geq 5$ ).

**10.5. General DS primitivity.** Theorem 9.2 proves gravitational coproduct primitivity for the principal reduction  $\widehat{\mathfrak{sl}}_2 \rightarrow \text{Vir}$ . The ghost-number obstruction argument uses only the universal grading of the BRST complex, and should extend to all principal reductions  $\widehat{\mathfrak{g}} \rightarrow \mathcal{W}(\mathfrak{g})$  (Conjecture ??). Non-principal reductions require a separate analysis: the ghost complex has a finer grading, and the transferred coproduct may be nontrivial.

**10.6. The descent fan.** The MC element  $\Theta_{\mathcal{A}}$  admits three independent projections with distinct targets and distinct symmetries:

- (i) *Categorical.*  $\mathcal{A} \rightarrow \bar{B}^{\text{ch}}(\mathcal{A}) \rightarrow \mathcal{A}^\dagger$ . The Koszul involution  $\mathcal{A} \leftrightarrow \mathcal{A}^\dagger$ .
- (ii) *Spectral.*  $\mathcal{A} \rightarrow \{S_r\} \rightarrow \sqrt{Q_L} \rightarrow \varepsilon_{Q_L}(s)$ . The functional equation  $s \leftrightarrow 1 - s$ .
- (iii) *Modular.*  $\mathcal{A} \rightarrow Z_g(\mathcal{A}) \rightarrow S_{\mathcal{A}}(v)$ . The Galois involution on the  $L$ -function content.

These three projections share  $\Theta_{\mathcal{A}}$  as common source but do not compose into a single functor. The Koszul, functional-equation, and Galois involutions are three different involutions on three different spaces; no functor intertwines all three. For lattice VOAs, all three projections are compatible (the lattice fan is closed). For class  $\mathbf{M}$  algebras, the compatibility is open.

The descent fan is the shadow-tower analogue of the trichotomy between algebraic, analytic, and arithmetic geometry: the categorical projection sees the algebra, the spectral projection sees the analysis, and the modular projection sees the arithmetic. Each is a faithful shadow of  $\Theta_{\mathcal{A}}$ ; none is complete.

**10.7. The genus-1 amplitude Mellin transform is Eisenstein.** The genus-1 degree-2 shadow amplitude has Fourier coefficients  $a_n = -24\kappa \sigma_1(n)$ , and the associated Dirichlet series satisfies  $D_2(\mathcal{A}, s) = -24\kappa \cdot \zeta(s) \zeta(s-1)$ : it is an Eisenstein  $L$ -function, not cuspidal. (The shadow  $L$ -function  $L_{\mathcal{A}}^{\text{sh}}(s) := \sum_{r \geq 2} S_r r^{-s}$ , which extracts degree-level constant terms, is a *different* object that does not equal  $-\kappa \zeta(s) \zeta(s-1)$ .) All connections between the shadow obstruction tower and Riemann zeta zeros are therefore mediated through Eisenstein scattering, not through direct spectral identification. This is both a negative result (the shadow does not “see” individual zeta zeros) and a positive one (the Eisenstein factorisation is a precise structural theorem).

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